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Institute of Computer Science
Chair of Communication Networks
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Part 2. Energy Efficiency Metrics and Models

NOMS 2026, Friday, 22 May 2026, 9:00 am - 12:30 pm

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https://comnets.org

SUSTAINET
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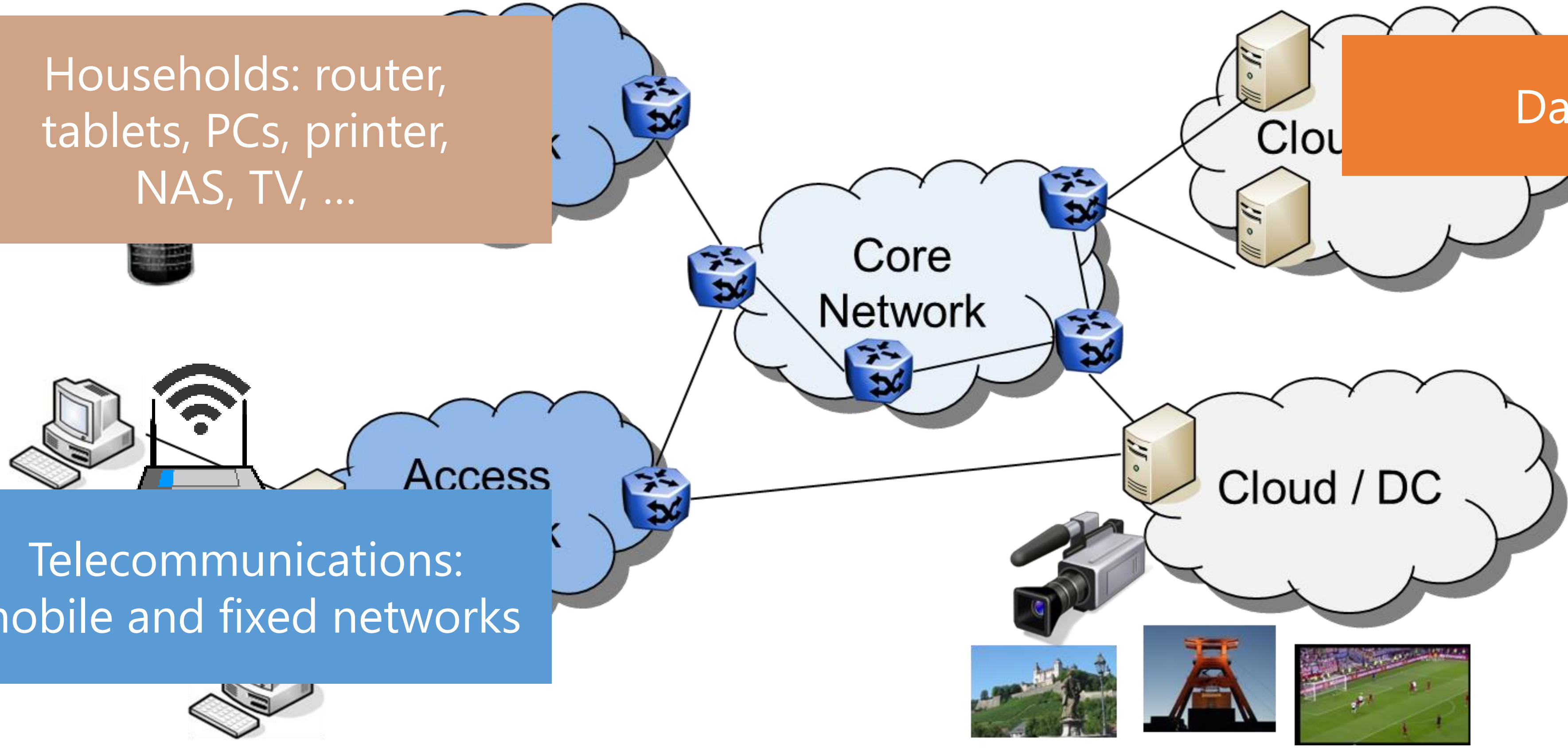
Contents (Part 2): Energy Efficiency Metrics and Models

- Energy efficiency metrics and classification
  - relevant energy efficiency metrics from literature, including widely used energy intensity
  - metrics defined by 3GPP and others for mobile communication systems
  - based on concrete examples and use cases
  - discussion and models along the metrics

- Energy modeling: assessing a system's energy efficiency
  - state-based and Markov reward model approaches
  - trade-offs between energy consumption and others like QoS

- Featured use case
  - mobile communication systems: from 5G/6G base stations to systems
  - beyond energy consumption and efficiency in 5G/6G: when reducing energy consumption may negatively affect overall sustainability

ICT Energy Consumption



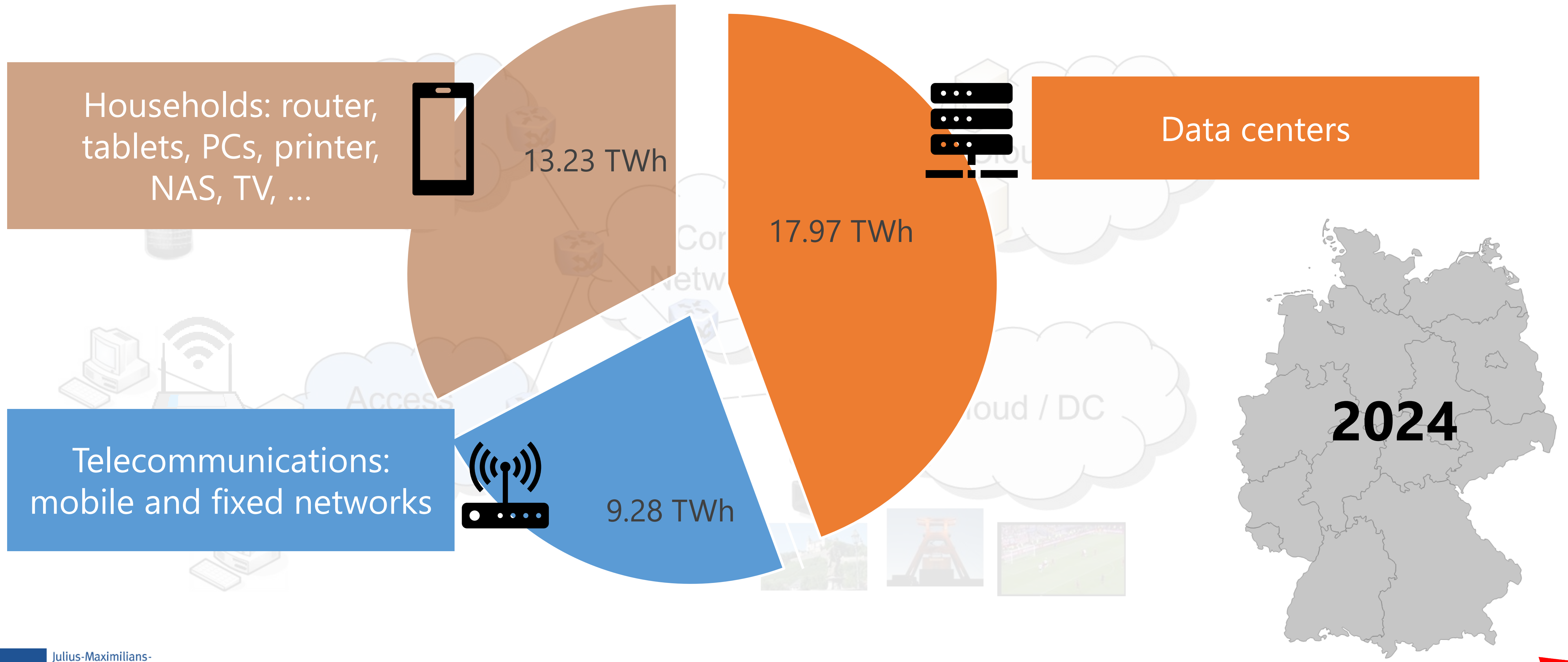
Households: router, tablets, PCs, printer, NAS, TV, …

Data Centers

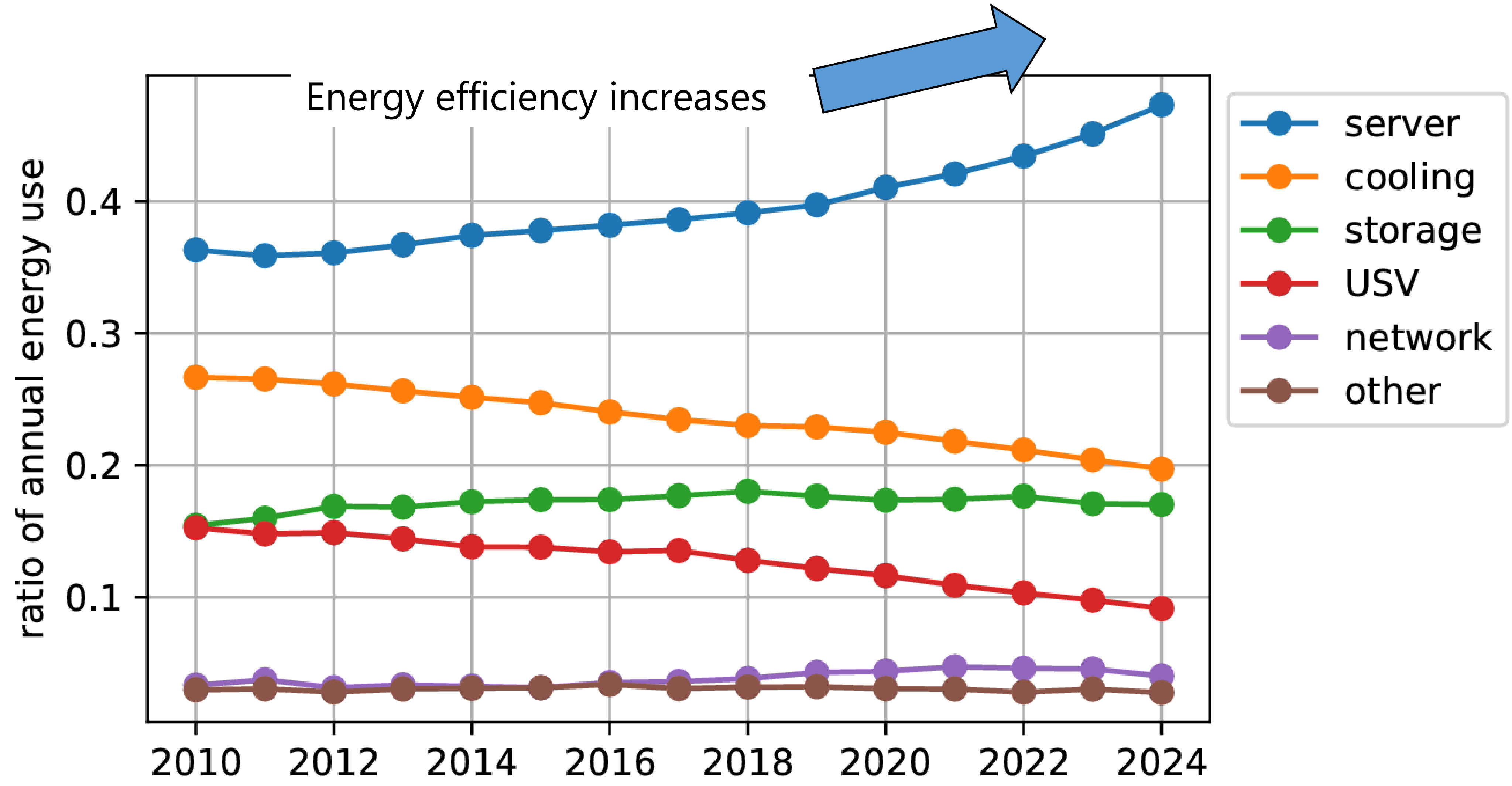
Telecommunications: mobile and fixed networks

ICT Energy Consumption in Germany in 2024 (Fraunhofer Study)

- Figures for operations only (excluding production) from 2024



Numbers on the Energy Use in Data Centers in Germany

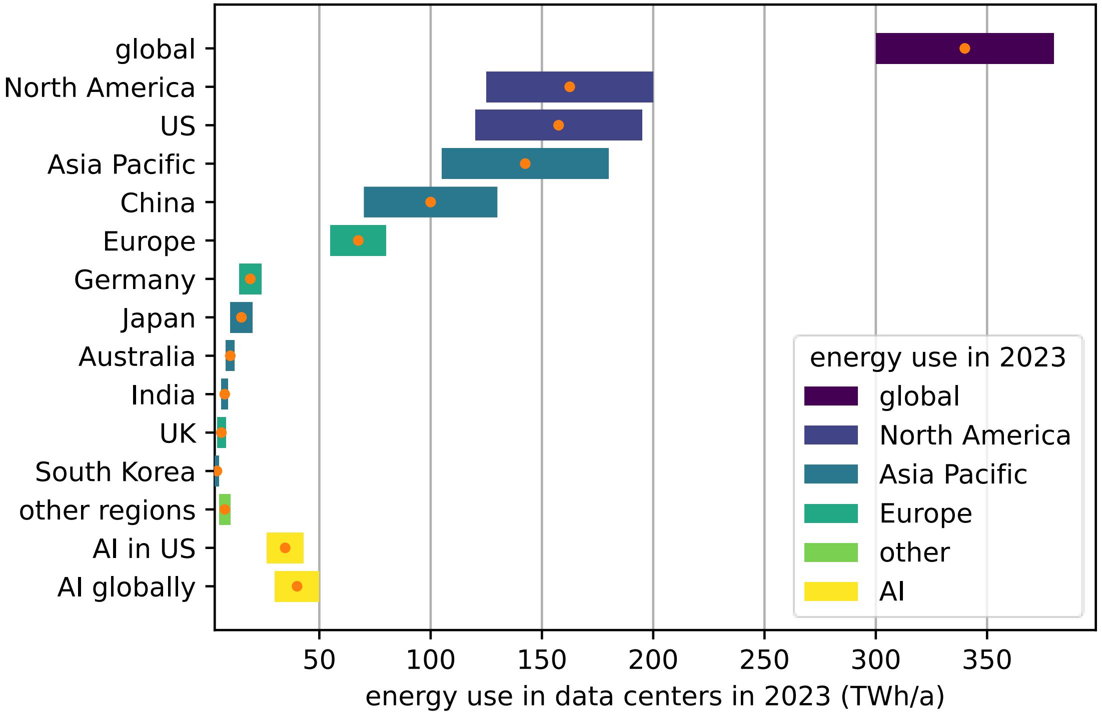


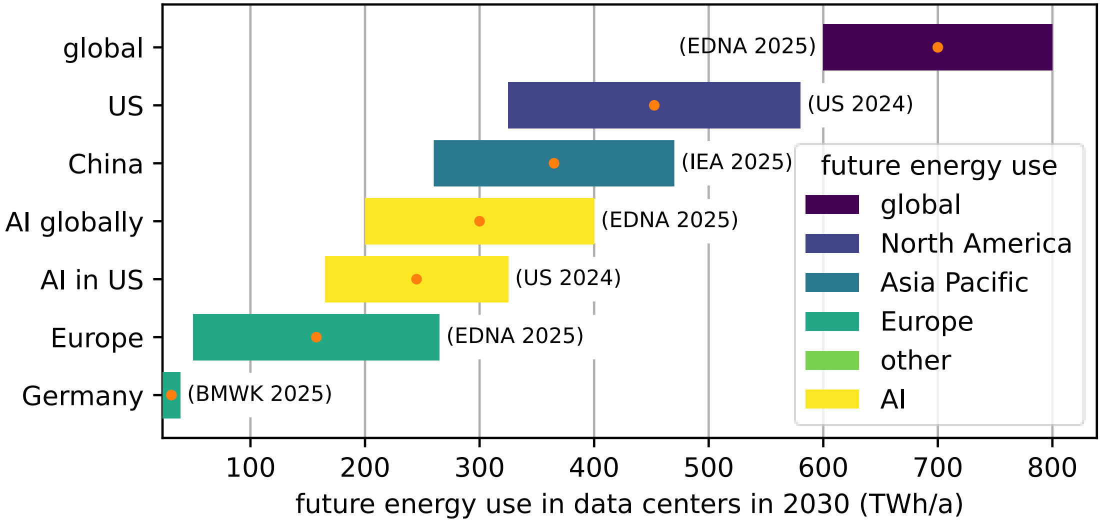
Hoßfeld, Tobias, "Energy Use in Data Centers: Current Figures and Trends", ITG News 03/2025, https://doi.org/10.25972/OPUS-41670

Example: Energy Use of Data Centers

- Increased energy use of data centers
- **But also increased workload, potentially increased value**

How to quantify energy efficiency?

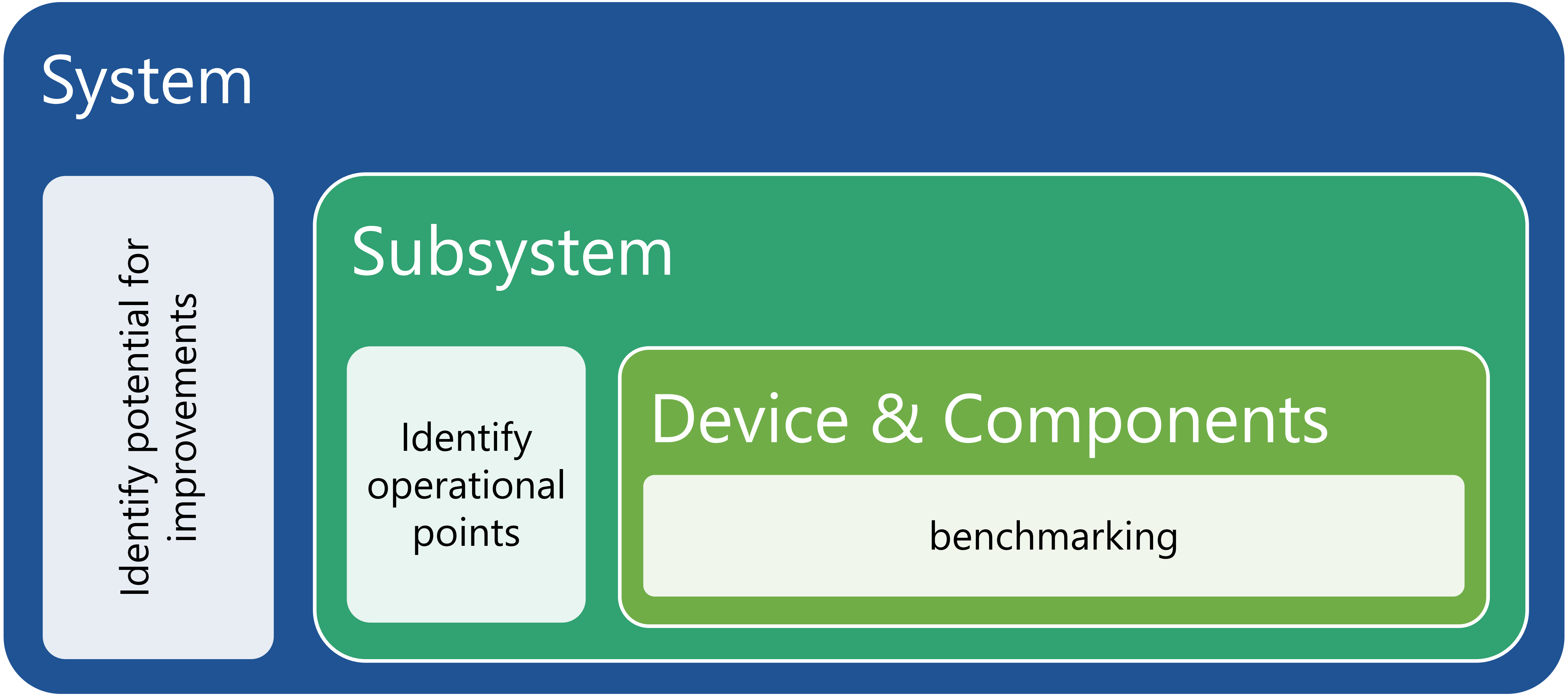




Hoßfeld, Tobias, "Energy Use in Data Centers: Current Figures and Trends", ITG News 03/2025, https://doi.org/10.25972/OPUS-41670

Classification of EE Metrics

Goal of Energy Efficiency Metrics



Example: SUSTAINET-Advance Project

- Sustainable Technologies for Advanced Resilient and **Energy-Efficient Networks** (SUSTAINET), CELTIC-NEXT project. https://www.celticnext.eu/project-sustainet/

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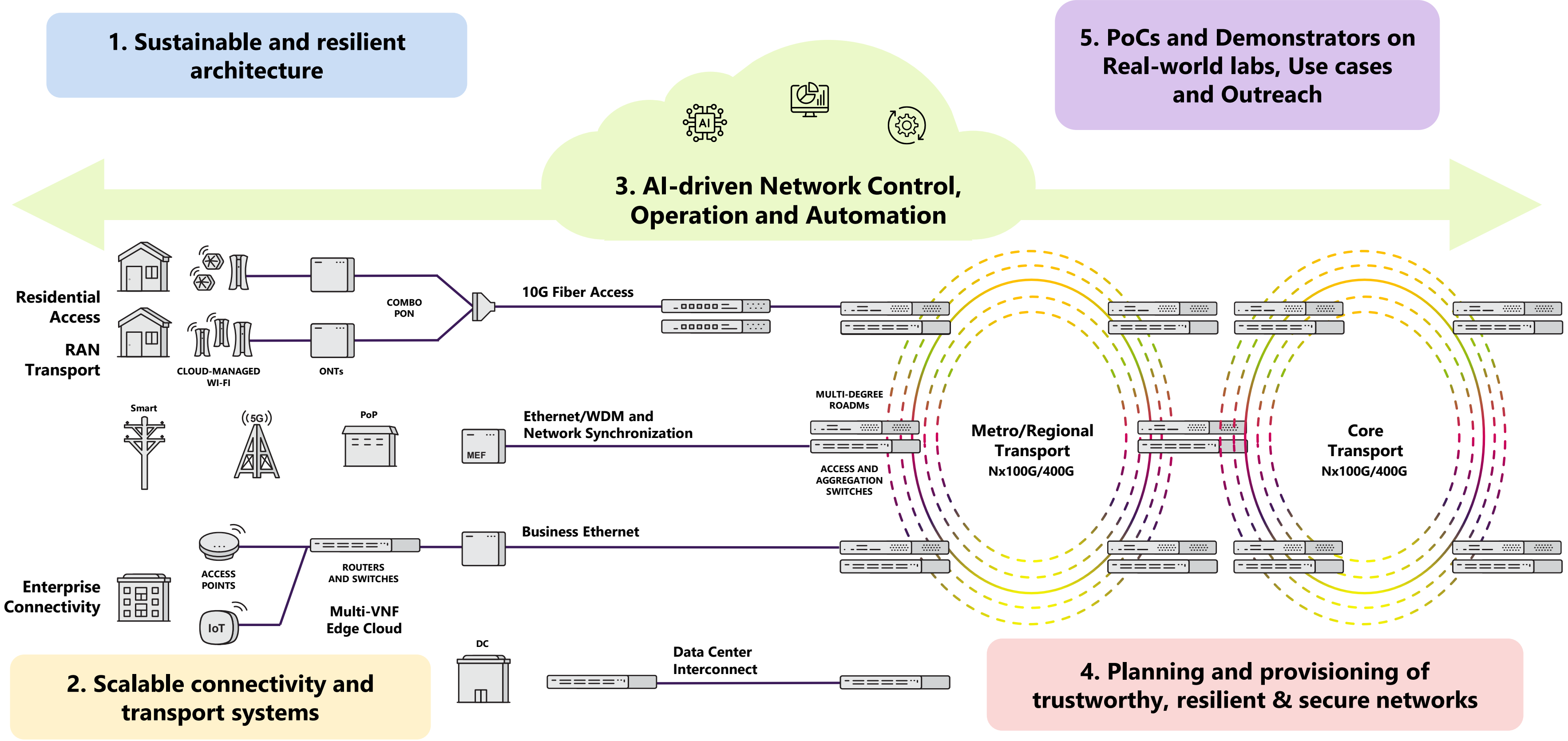


Energy Efficiency Framework

Input: energy or power consumption for IT equipment, servers, networking, and supporting infrastructure

Output: computational work, networking load, data and traffic processed, services delivered, utilization, load

Energy efficiency
(of system, subsystem, device, component)

Efficiency

- Efficiency signifies the level of performance that uses the least amount of inputs to achieve the highest amount of output

- In general, efficiency is a **measurable** concept, quantitatively determined by the ratio of useful output to total useful input
  - $r = G/B$, where G is the amount of useful output ("product“ or “goodness”) produced per the amount B ("cost“ or “badness”) of resources consumed
  - for example: bit-per-Joule efficieny

- Efficiency often expressed as **percentage of the result that could ideally be expected**
  - for example if no energy were lost due to friction or other causes, in which case 100% of fuel or other input would be used to produce the desired result

https://en.wikipedia.org/wiki/Efficiency

Classification of EE Metrics

Input: energy or power consumption for IT equipment, servers, networking, and supporting infrastructure

Energy Efficiency

Absolute Measures: Consumption Related to Output

Relative Measures (%): Relation within System

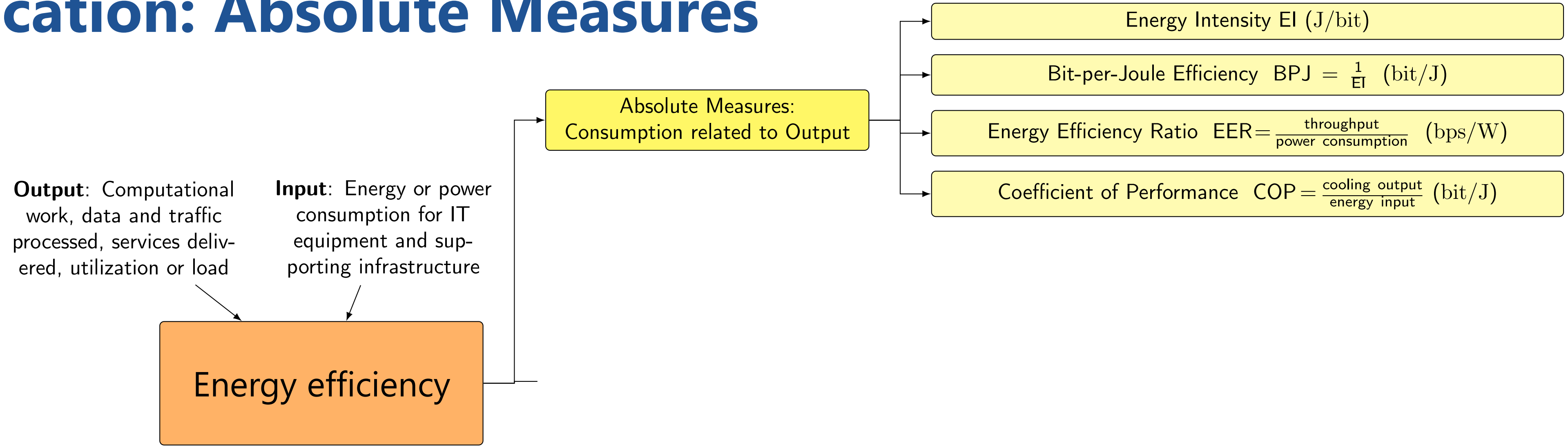
Relative Measures (%): (Ideal) Reference System

Output: computational work, networking load, data and traffic processed, services delivered, utilization, load

Hossfeld, T., Wunderer, S., Loh, F., & Schien, D. (2024). Analysis of Energy Intensity and Generic Energy Efficiency Metrics in Communication Networks: Limits, Practical Applications and Case Studies. IEEE Access. https://doi.org/10.1109/ACCESS.2024.3435716

Absolute Measures: Energy Intensity

Classification: Absolute Measures



- The **energy intensity (EI)** of a system is the ratio of the energy consumption $y = f(x)$ and the traffic volume x processed by the system over the same time frame.

$$I(x) = \frac{f(x)}{x}$$

- The **bit-per-joule efficiency (BPJ)** of a system is the ratio of the traffic volume x and the energy consumption y over the same time frame Δt. Also known as **Energy Efficiency ratio (EER)**.

$$\gamma(x) = \frac{x}{f(x)} = \frac{x/\Delta t}{f(x)/\Delta t} = \frac{1}{I(x)}$$

Example: Energy Intensity (EI)

- *Example*: The energy intensity of a system is 5 Joule / Mbit.

How much power does it consume for 100 Mbps?

(a) 500 Joule

(b) 50 Joule

(c) 5 Joule

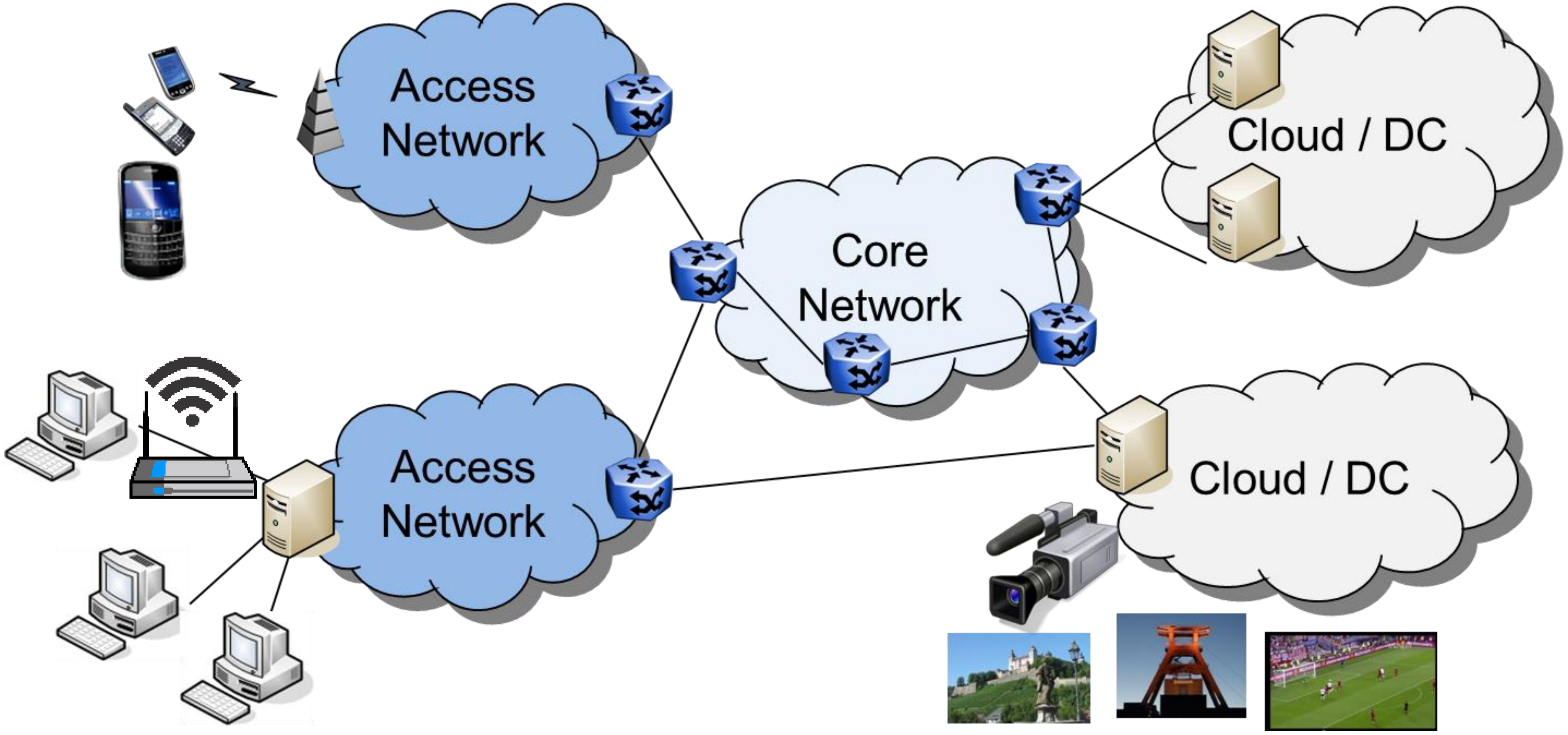
- A car's fuel consumption is 10 liters per 100 km when there is one passenger on board.

What is the fuel consumption when there are 5 passengers?

Example: Mobile Base Stations

Mobile Networks Energy Consumption (GSMA 2026)

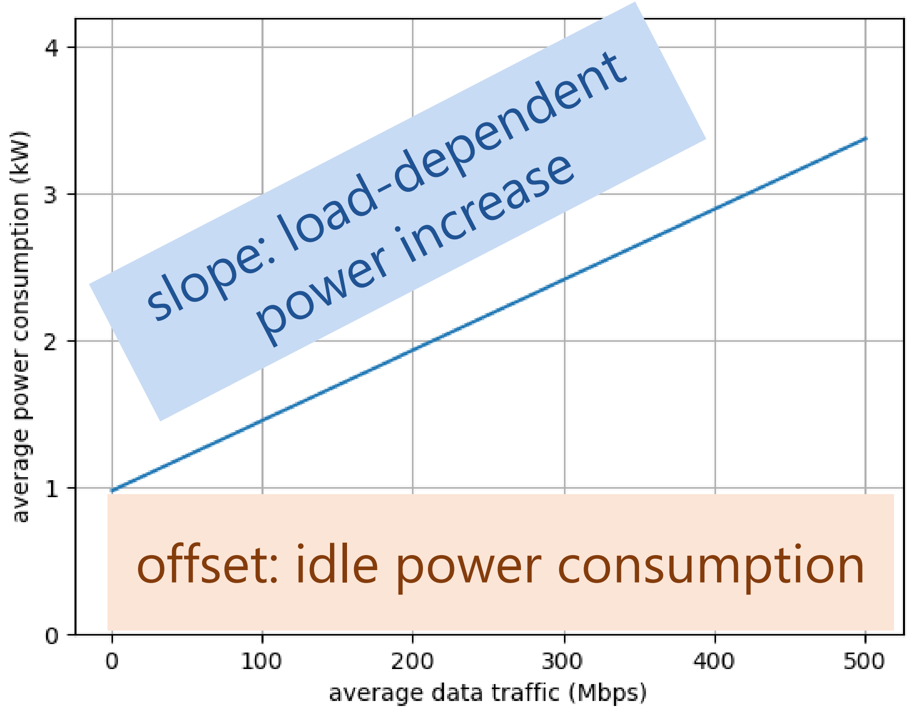
RAN 87% | **Core** | **Data Centers**



https://www.gsmaintelligence.com/research/going-green-measuring-the-energy-efficiency-of-mobile-networks-and-towers

Example: Linear Power Model for 5G Base Stations

- Measurements of single 5G 1-band base station in dense urban area
- Power model: 5G 1-band, dense urban
  - Carrier frequency 3.5 GHz
  - Duplexing: TDD 3:1
  - Bandwidth 50 MHz
  - 3 Sectors
  - Population density in [1500; 10000[with an average of 2925 citizens/km^2
  - Average area of 1.37 km^2 with standard deviation 0.99 km^2
- And more linear models in literature …



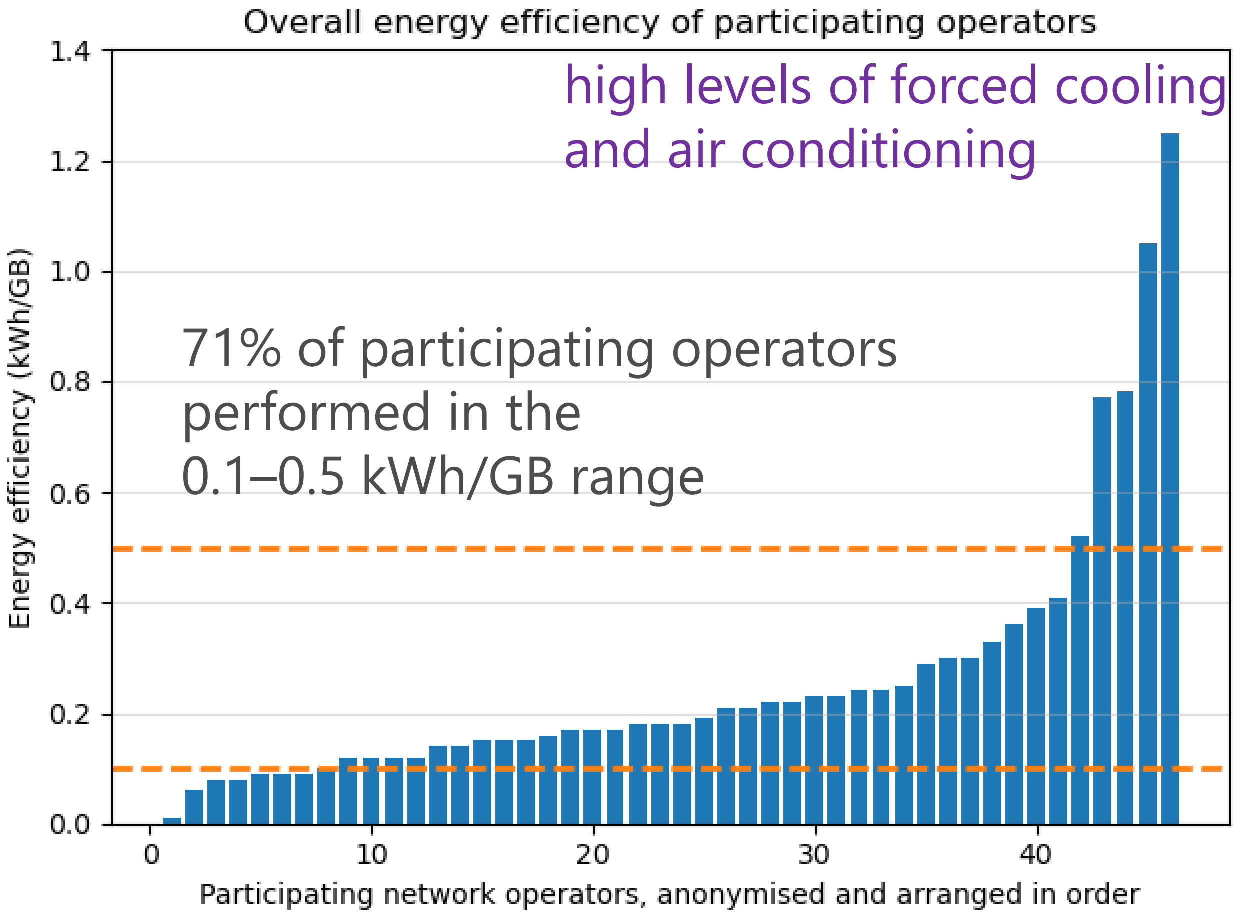
Golard, L., Louveaux, J., & Bol, D. (2023). Evaluation and projection of 4G and 5G RAN energy footprints: The case of Belgium for 2020–2025. *Annals of Telecommunications*, *78*(5), 313-327.

M. Oikonomakou, A. Khlass, D. Laselva, M. Lauridsen, M. Deghel, and G. Bhatti, "A power consumption model and energy saving techniques for 5G-advanced base stations," in *IEEE ICC Workshop GreenNet 2023*.

GSMA Report 2026: Energy Efficiency of Mobile Networks

- **Energy consumption**
  - 87% of energy is consumed in the radio access network (RAN).
  - 11% network core and operator-owned data centers
  - 2% other operations (includes offices, retail and logistics)

- Energy efficiency in terms of **energy intensity**
  - An average of 0.08 kWh of energy to transfer 1 GB of data across their RAN.
  - Down from 0.10 kWh per GB in 2024 – a slight but still notable improvement.



Data taken from: Kolta, E., & Hatt, T. (2026). Going green: measuring the energy efficiency of mobile networks. GSMA Intelligence, Tech. Rep.

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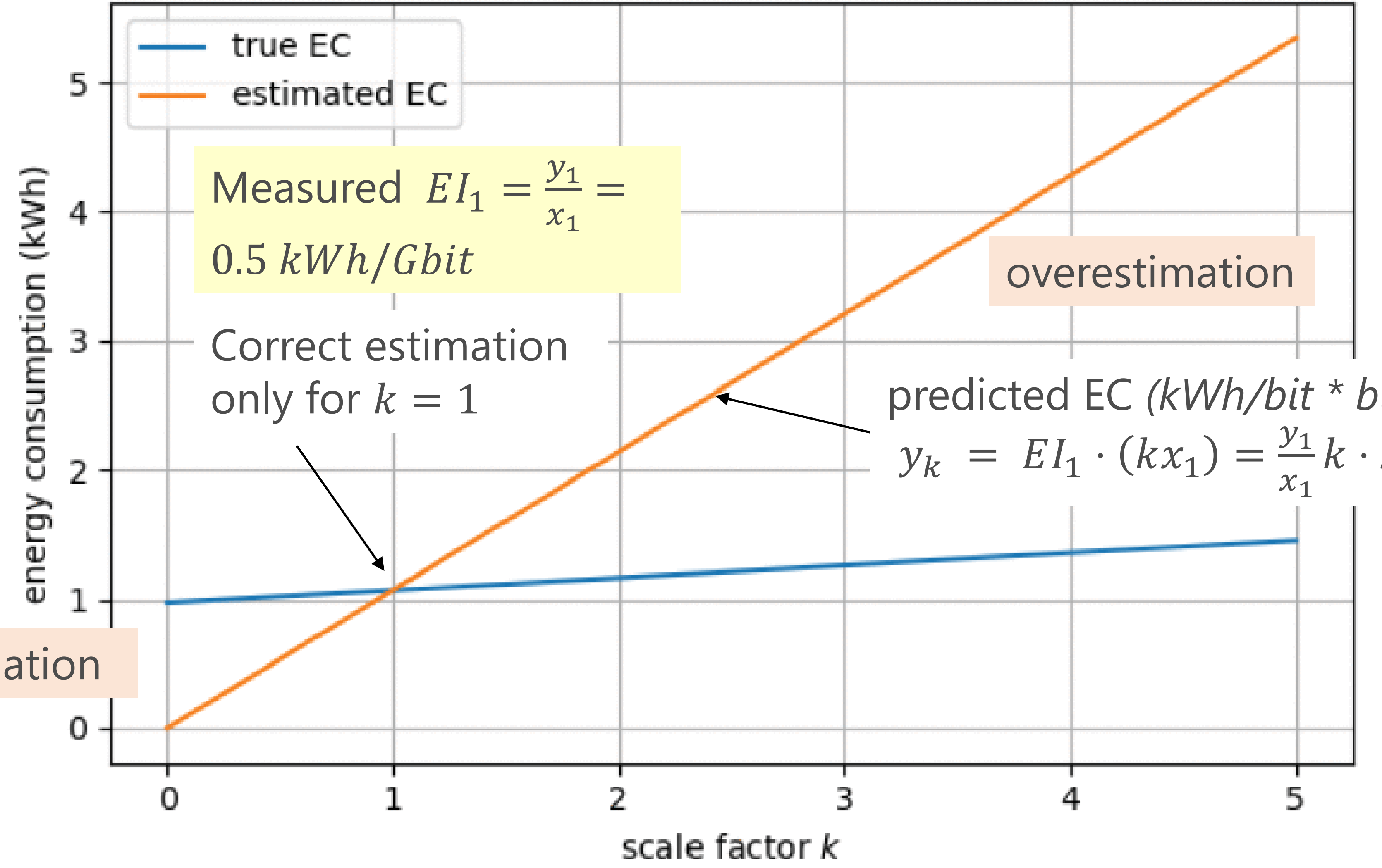
EI: Prediction Error based on Measured EI

- Linear 5G base station model over one hour, depending on the traffic scale factor

Measured $EI_1 = \frac{y_1}{x_1} = 0.5\ kWh/Gbit$

EI: Prediction Error based on Measured EI

- Linear 5G base station model over one hour, depending on the traffic scale factor



Measured $EI_1 = \frac{y_1}{x_1} = 0.5\ kWh/Gbit$

overestimation

Correct estimation only for $k = 1$

predicted EC *(kWh/bit * bit)*

$$y_k = EI_1 \cdot (kx_1) = \frac{y_1}{x_1} k \cdot x_1 = y_1 \cdot k$$

underestimation

Energy Intensity Improvements by Increased Traffic

- Increased traffic results in better utilization of resources ➔ better energy efficiency and EI

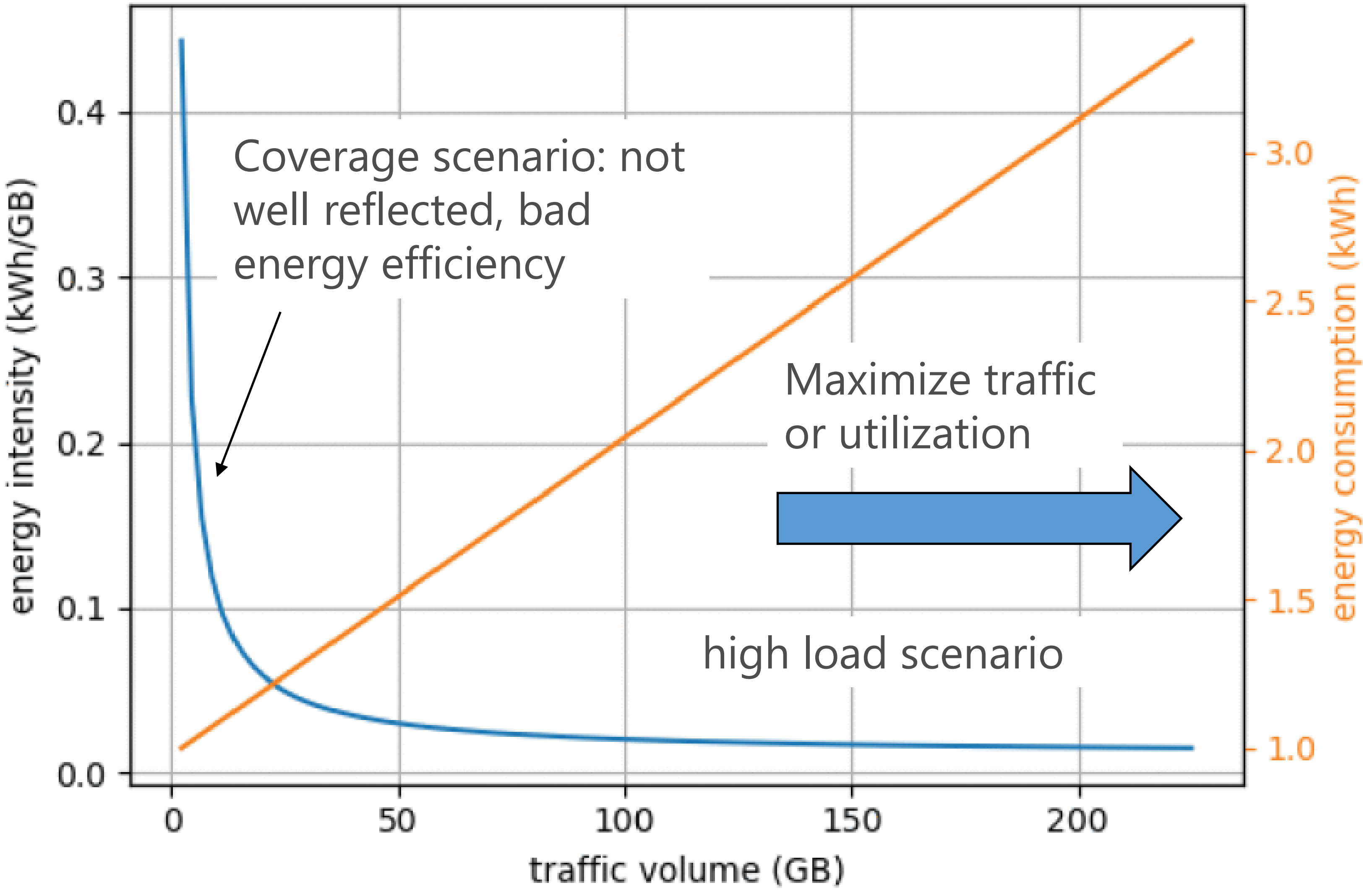
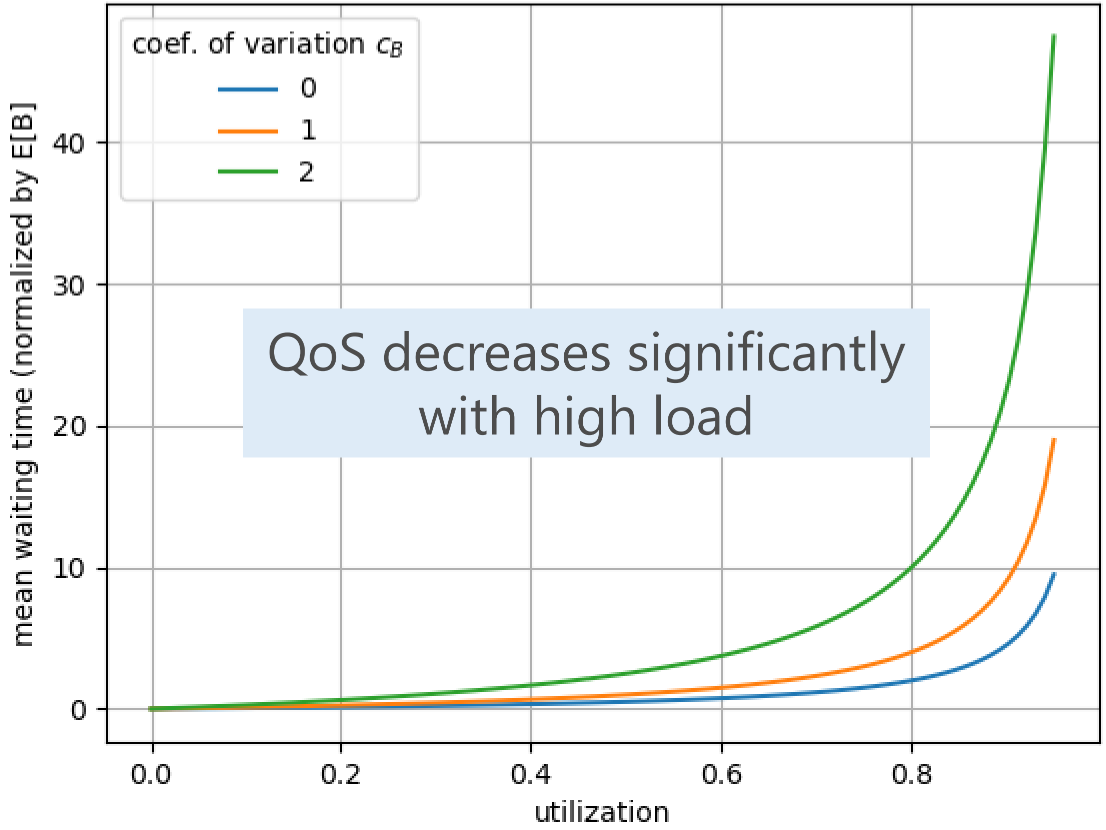
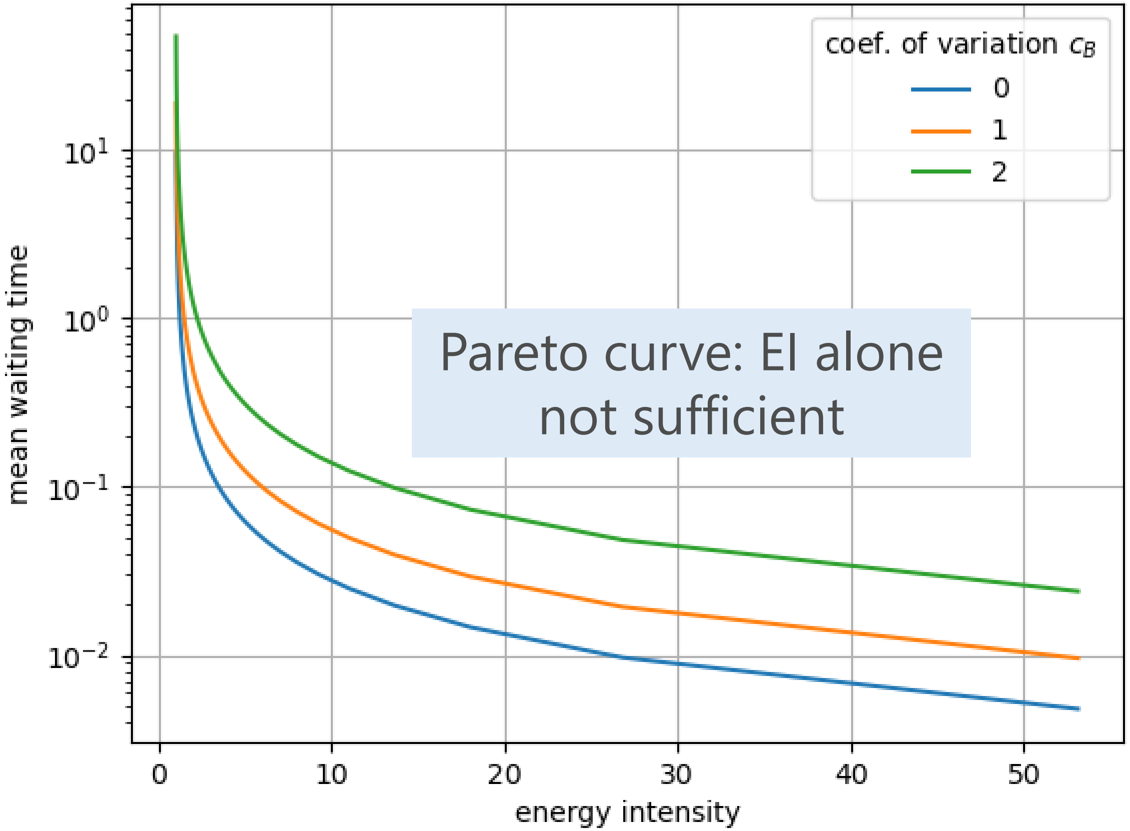


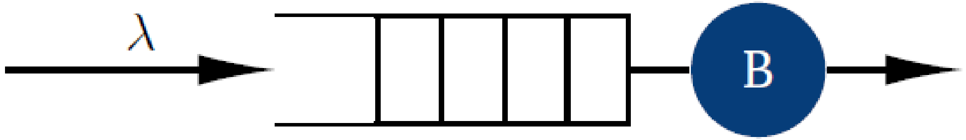
Illustration: Single Waiting Queue



QoS decreases significantly with high load



Pareto curve: EI alone not sufficient

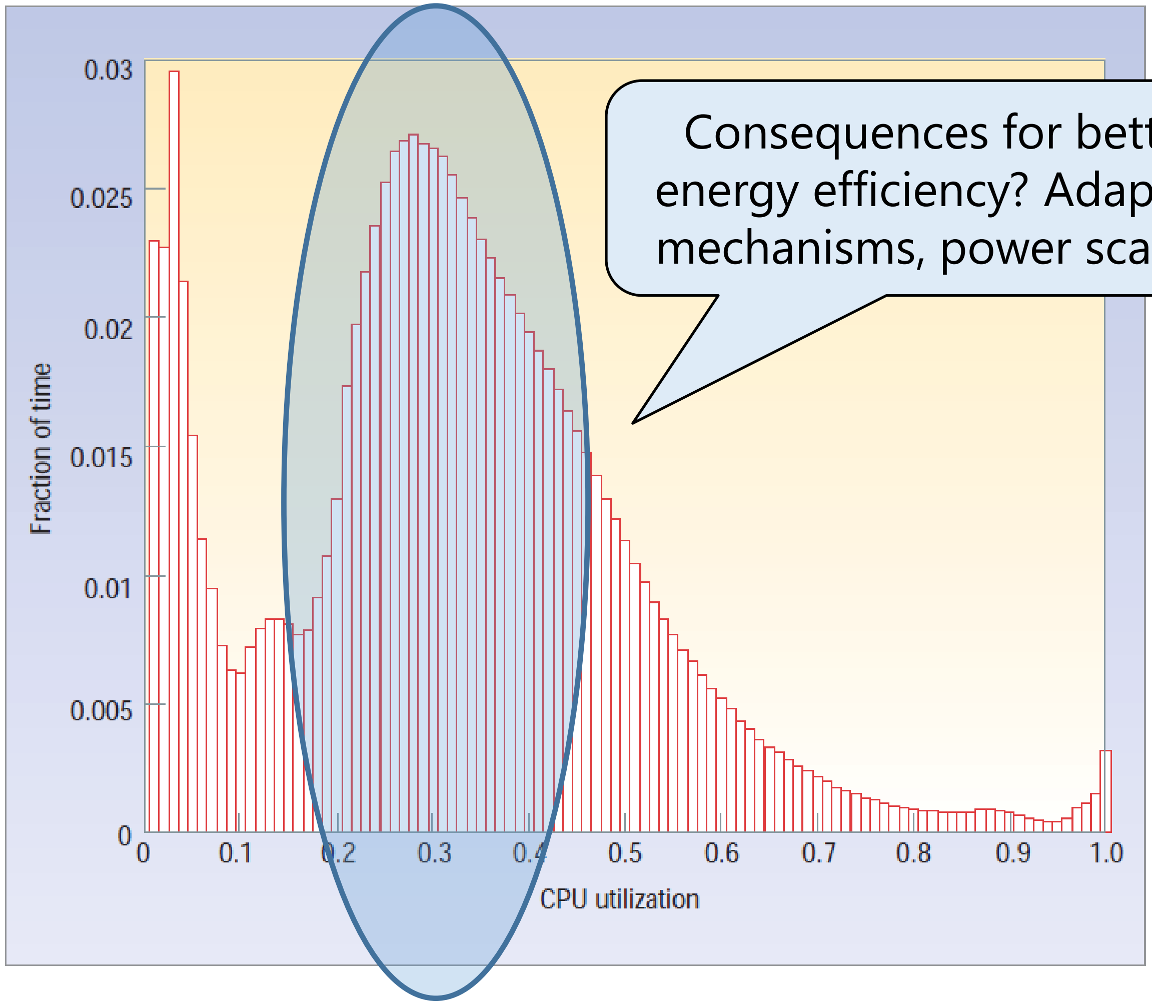


Measurements of Server Utilization

- Average CPU utilization of more than 5,000 servers during a six-month period.
- Servers are operating most of the time at between 10 and 50 percent of their maximum utilization levels.

Why such a peak at 30%?

see later utilization-based EE metrics



Consequences for better energy efficiency? Adaptive mechanisms, power scaling

Barroso, L. A., & Hölzle, U. (2007). The case for energy-proportional computing. *Computer*, 40(12), 33-37.

Energy Intensity: Most Common Metric … but

- Difficult to compare across heterogeneous systems and workloads

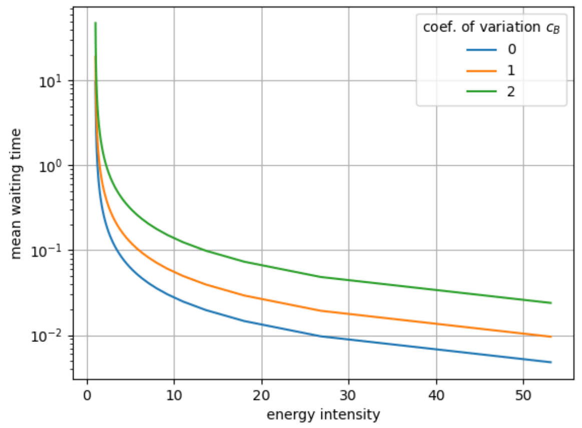
- Prediction based on a particular EI are typically wrong
  - constant offset for devices and systems
  - ignores system behavior, e.g. switching on/off (additional) devices

- Sensitive to traffic load and utilization conditions

- Can hide trade-offs with QoS or latency
  - lowering transmit power may improve energy intensity → but can increase packet loss
  - optimizing purely for J/bit may favor high utilization→ but overloaded systems can degrade QoS and user experience.

- Optimizing one metric may worsen overall sustainability

EE Metrics: QoS and SLAs



Energy-Delay Product

- **Energy-delay product (EDP):** $\mathbf{EDP} = \boldsymbol{D \cdot E}$
  - Combines energy consumption and performance in a single metric
  - Captures the trade-off between energy consumption E and delay D
  - Lower EDP indicates a more energy-efficient and faster system
  - Highlights that minimizing energy alone may increase latency, and vice versa
- Absolute values difficult to interpret, may be used to compare systems

- A **weighted sum** may achieve the same, but gives more control on the importance of factors: $\boldsymbol{S = w_D \cdot D + w_E \cdot E}$

Mode	Energy per hour	Average delay	EDP
Always active	10 kWh	5 ms	50 kWh·ms
Light sleep	6 kWh	10 ms	60 kWh·ms
Adaptive sleep	4 kWh	15 ms	60 kWh·ms
Deep sleep	2 kWh	50 ms	100 kWh·ms

Useful Work and SLAs: ESV

- Energy-saving mechanisms can degrade service quality.
  - For example: higher latency or packet losses when waking up devices or when remaining resources are overloaded
  - **Service Level Agreement violations** indicate when system performance no longer meets required QoS targets (which may be adjusted to **services**)
  - Minimizing energy alone may encourage overly aggressive sleep modes or resource shutdowns.

- Composite metric taking into account energy consumption and SLA violations rate:

$$ESV = energy \cdot SLAV$$

  - Key idea: a system is only "efficient" if it still delivers acceptable service quality
  - Lower ESV indicates a better energy-quality trade-off
  - Difficult to interpret

Mode	Energy	SLA violations	ESV
Always active	10 kWh	1%	0.10
Adaptive	5 kWh	3%	0.15
Aggressive saving	2 kWh	15%	0.30

Pertric: Taking into account Host Shutdowns

- In practice, number of host shutdowns (NHS) matter
  - reduce hardware lifetime,
  - increase maintenance costs,
  - cause instability and recovery delays.
- **Pertric** (*Per*formance Me*tric*): objective is to maximize overall performance while minimizing energy consumption, average SLA violations, and the number of host reactivations.

$$\boldsymbol{Pertric = energy \cdot SLAV \cdot NHS}$$

with number of host shutdowns (NHS), SLA violation rate (SLAV), energy consumption

- Example: Including SLA violations and host shutdowns changes the ranking

Strategy	EC	Data	EI	SLAV	NHS	ESV	Pertic
Conservative	10	20	0.8	0.01	1	**0.10**	0.30
Adaptive	5	50	0.5	0.03	2	0.15	**0.10**
Aggressive	**2**	100	**0.2**	0.10	5	0.20	0.50

Discussion: Is it reasonable to mix the metrics?

- Advantages
  - Capture trade-offs between energy, QoS, and operational stability
  - Simplify optimization and comparison of system strategies
  - Reflect practical engineering objectives more realistically.

- Challenges and Pitfalls
  - Different dimensions are mixed into a single number
  - Weighting and scaling strongly influence conclusions
  - Results may become difficult to interpret physically
  - Important information can be hidden by aggregation.

- Alternative: use separate metrics for
  - energy efficiency
  - QoS/SLA performance
  - reliability/stability.

clearer interpretation,
Pareto analysis,
transparent trade-off discussions

Mobile Network Specific Metrics

Absolute Measures

Purpose of the EE Metrics

- **Comparison**: Is MNO 1 better than MNO 2? Is configuration 1 better than config. 2?
- **Benchmark**: How efficient is the MNO (against industry best practices or internal targets)?
- **Planning**: Is it worth upgrading the network? Does it make sense to get another/new device?
- **Operations**: How to reduce operational inefficiencies? What is the impact of changes? Are there problem areas?
- **Reporting**: What’s the MNO's environmental footprint (also in terms of energy consumption)?

with respect to a particular EE metric of interest …

… but also quality, costs, …

ITU-T Rec. L.1331 Assessment of mobile network energy efficiency

ITU-T Rec.
L.1331

Defines energy efficiency metrics and assessment methods for mobile networks.

operational energy consumption metrics (how much energy is used),

performance metrics (traffic, coverage, latency, users),

combines them into application-specific energy efficiency indicators.

Applicable from GSM to 5G NR systems.

covers network elements such as: base stations, backhaul systems, radio controllers.

Cooling systems, power supply systems, supporting infrastructure

Mobile network data energy efficiency

ITU-T Rec. L.1331

- **Mobile network data energy efficiency ($EE_{MN,DV}$):** ratio between the data volume (DV_{MN}) and the energy consumption (EC_{MN}) when assessed during the same time period.

$$EE_{MN,DV} = \frac{DV_{MN}}{EC_{MN}}$$

where $EE_{MN,DV}$ is expressed in bit/J. This is the bit-per-Joule efficiency.

- **Operational energy** consumed by the assessed mobile (sub-)network: base stations, backhaul equipment, site infrastructure, control nodes, cooling systems, power supply, …
- **Not included**: embodied/manufacturing energy, device/user equipment energy, construction and lifecycle emissions.
- **Data volume** represents useful traffic delivered by the network. Usually excludes signaling traffic.
- **Time duration** of measurements shall be: one week, one month, or one year

QoS: Latency based metric

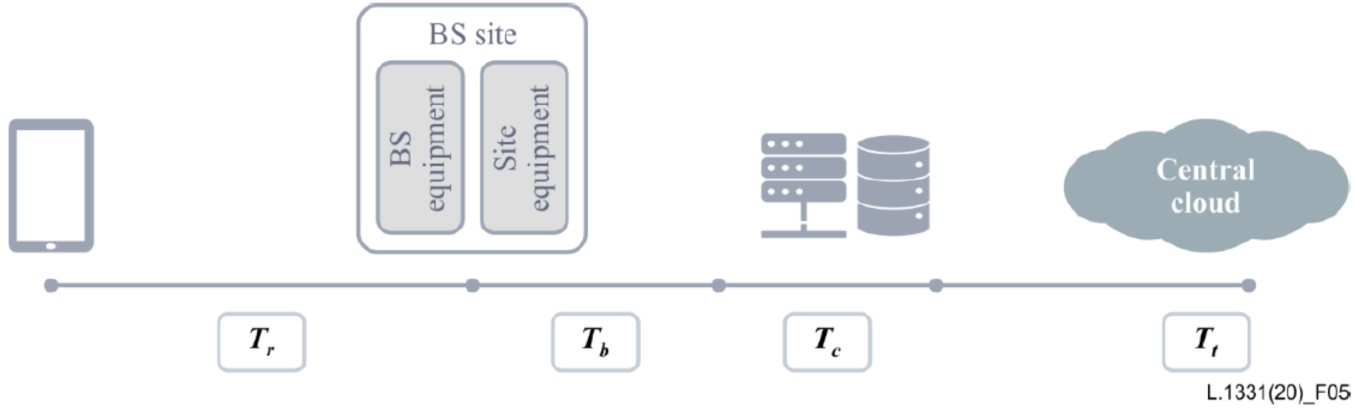
ITU-T Rec. L.1331

- **Latency based metric ($EE_{MN,L}$):** the inverse ratio of the end-to-end user plane latency $T_{e2e,MN}$ and the energy consumed EC_{MN} by the MN:

$$EE_{MN,L} = \frac{1}{T_{e2e,MN} \cdot EC_{MN}}$$

where $EE_{MN,L}$ is expressed in ms^{-1}/J.

- Definition of the end-to-end user plane latency components



radio backhaul core transpot from core to Internet/cloud

figure taken from ITU-T Rec. L.1331

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QoS: Energy Efficiency for Network Slicing

ITU-T Rec. L.1331

- For networks with Network Slicing functionalities the EE is measured as follows, where the **performance P_{ns}** is related to the type of Network Slicing implemented

$$EE_{MN,ns} = \frac{P_{ns}}{EC_{ns}}$$

- **eMBB** (enhanced Mobile Broadband): high data rates and capacity for applications such as video streaming, AR/VR, and broadband access

$$P_{ns} = DV_{mn} \quad \Rightarrow \quad EE_{MN,eMBB} = EE_{MN,DV} = \frac{DV_{MN}}{EC_{MN}}$$

- **URLLC** (Ultra-Reliable Low-Latency Communication): ultra-low latency and high reliability for mission-critical applications such as industrial automation and autonomous driving.

$$P_{ns} = \frac{1}{T_{e2e,MN}} \quad \Rightarrow \quad EE_{MN,URLLC} = EE_{MN,L} = \frac{1}{T_{e2e,MN} \cdot EC_{MN}}$$

- **MMTC** (massive Machine-Type Communication): scalable connectivity for very large numbers of low-power IoT devices with small data transmissions.

$$P_{ns} = N_{MMTC} \quad \Rightarrow \quad EE_{MN,MMTC} = \frac{N_{MMTC}}{EC_{MN}}$$

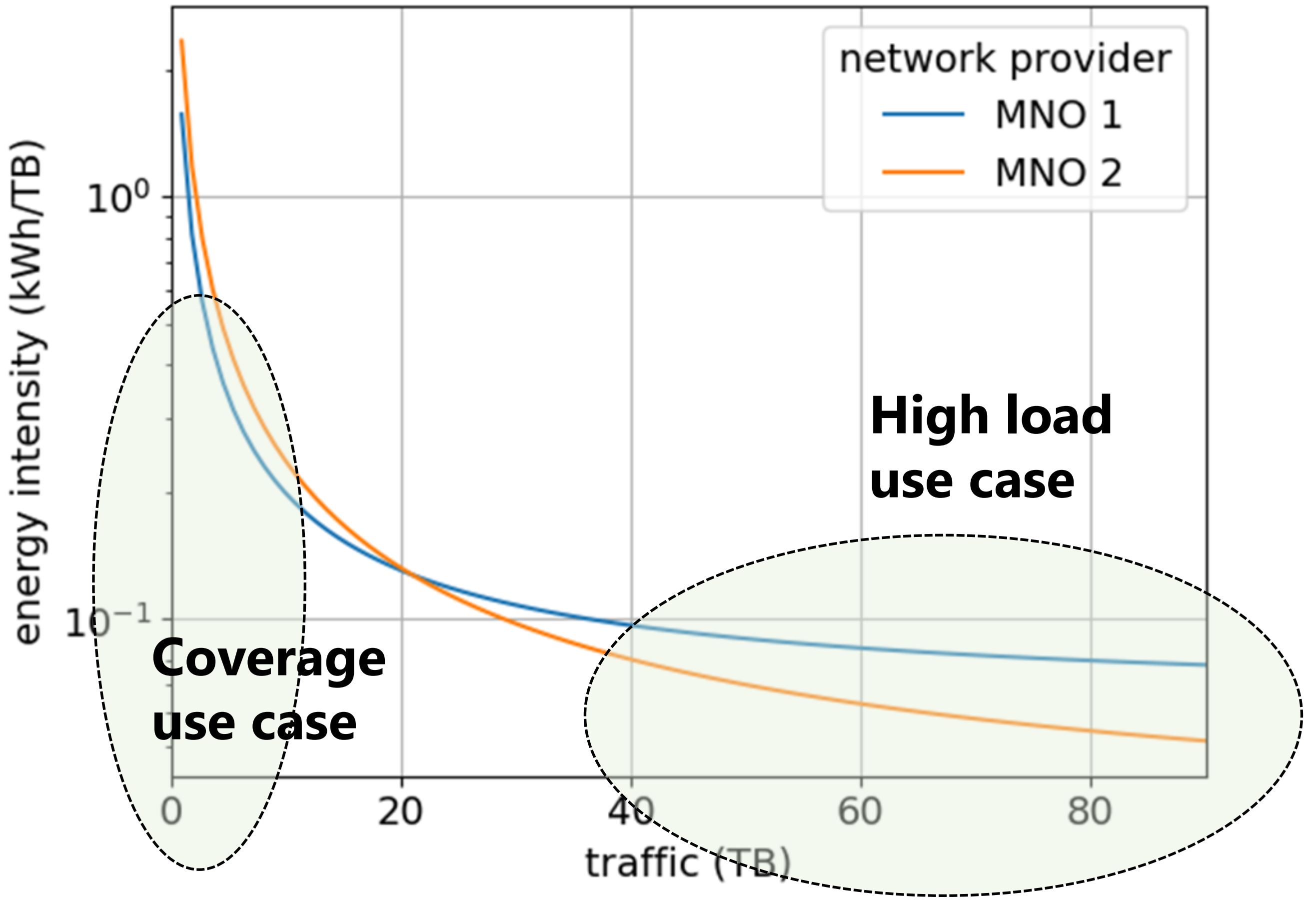
Coverage of a Single Base Station

- Just considering the traffic volume is not sufficient

- Mobile networks are offering more than bits only: services
  - coverage
  - mobility
  - quality of service

- Joule / Bit is misleading for the coverage use case
  - Poor EI values for low loads
  - Compared to high load scenarios

  ➔ EI not appropriate for all services



Mobile network coverage energy efficiency

ITU-T Rec. L.1331

- **Mobile network coverage energy efficiency (*$EE_{MN,CoA}$*)** is the ratio between the area covered by the MN and the EC when assessed for one year

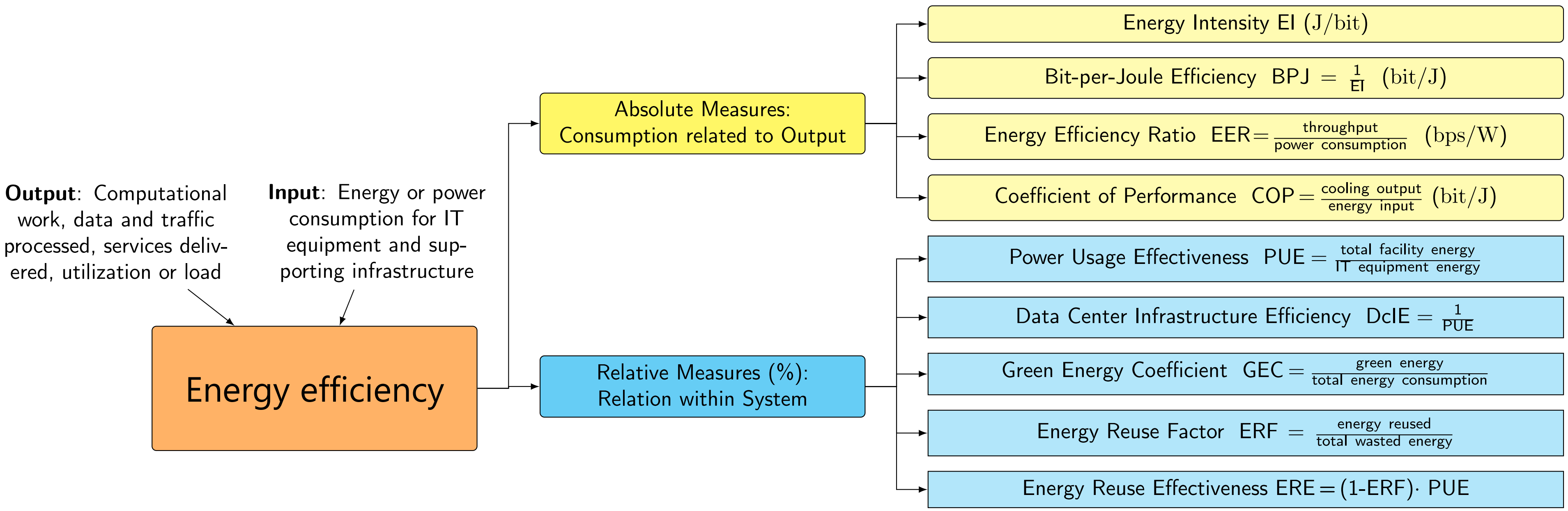
$$EE_{MN,CoA} = \frac{CoA\_des_{MN}}{EC_{MN}}$$

- $EE_{MN\ CoA}$ is expressed in m^2/J and EC_{MN} is the yearly energy consumption and CoA_des_{MN} is the "coverage area".

- **Geographic coverage area**: total area where the mobile network operator is licensed to provide service.
  - The licensed area may not be fully covered by the network.
  - Coverage obligations may include:
    - geographic coverage (e.g., >90% of country area)
    - population coverage (e.g., 98% of population served).
- **Designated coverage area** is the area to be covered based on network planning
- Coverage quality: how well the users within the coverage area are covered (SINR)

Relative EE Measures: Relation within System

Classification: Relative EE Measures – Relation within System



- Power Usage Effectiveness **PUE** = total facility energy use / IT equip. energy use
  - Focuses on efficiency of cooling infrastructure, not on energy efficient system
- Green Energy Coefficient **GEC** = green energy / total energy use

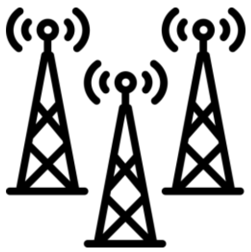
Power Usage Effectiveness (PUE)

- PUE is a metric used to evaluate data center energy efficiency.
  - Ratio of total power consumed by a data center (including IT equipment and infrastructure) to the power consumed by IT equipment alone.
  - Lower PUE values indicate higher energy efficiency.

- Energy Requirement of Data Centers in Germany (2022)
  - Energy consumption increased from 10.5 TWh/a to 16 TWh/a
  - Workload increased by a factor of 8 from 2010 to 2020
  - PUE in 2010: 1.98
  - PUE in 2020: 1.63
  - Increase of PUE: 1.98/1.63 = 121%
  - EI in 2010: 10.5 TWh/a / Workload
  - EI in 2020: 16 TWh/a / (8·Workload) = 2 TWh/a / Workload
  - Increase of efficiency: 10.5/2 = 525%

Tower Energy Efficiency Index (TEEI)

- Power usage effective applied to the cell tower / base station site
- **Site power usage effectiveness (PUE),** also known as the **tower energy efficiency index (TEEI)**
  - relates the total energy consumption E_{total} of the site (tower / base station)
  - to the energy consumed by the telecom equipment E_{telco}

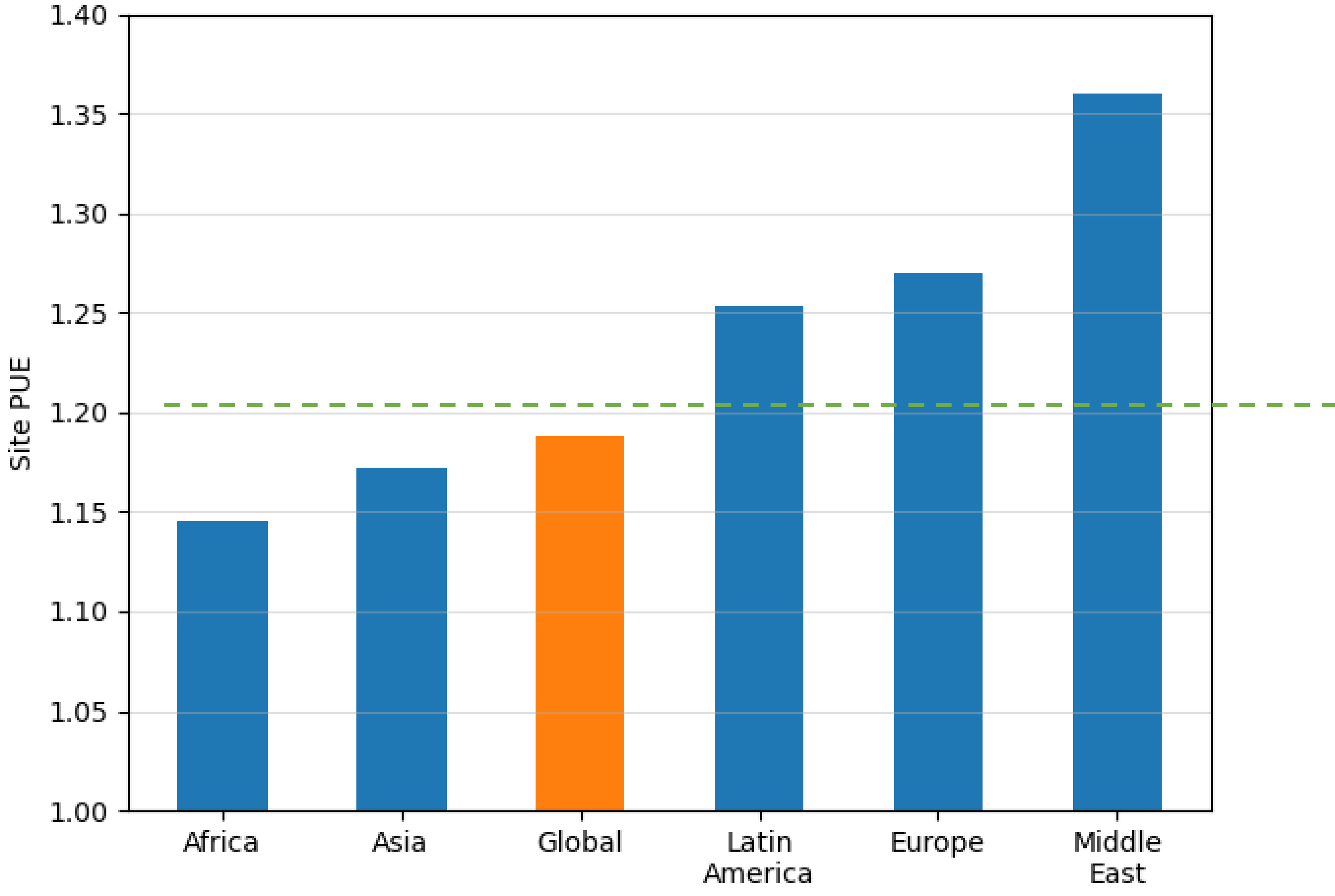


$$TEEI = \frac{E_{total}}{E_{telco}}$$

- Any value greater than 1 represents
  - extra, non-telecom energy consumed to run the site (e.g., cooling systems, lighting, and power loss through conversions)
  - compared to the energy actually used for connectivity
- *Note in general: Efficiency = 1/Effectiveness (which is not the case here)*

Site PUE levels for Providers (GSMA 2026)

- Data based on the GSMA Intelligence Energy Efficiency Analysis and Benchmarking project in 2026, with 118 networks.
- Towercos must adapt directly to whatever conditions exist around each physical site: temperature, vandalism, security, humidity, geography and regulatory environment
- Middle East's high PUE level is driven by high levels of forced cooling and air conditioning,
- Africa's low PUE level is likely driven by predominantly outdoor sites with no air-conditioning and simpler power systems



Source: GSMA 2026

Data Center Efficiency Rating

- Power Usage Effectiveness (PUE): Compare the energy consumption of computing applications and infrastructure equipment with that of IT equipment
  - $PUE = \frac{\text{Total facility energy}}{\text{IT equipment energy}}$
- Data Center infrastructure Efficiency (DCiE): Measure the percentage of power consumed within the data center.
  - $DCiE = 1/PUE = \frac{\text{IT equipment energy}}{\text{Total facility energy}}$

PUE	DCiE	Levels of efficiency
3.0	33%	very inefficient
2.5	40%	inefficient
2.0	50%	medium
1.5	67%	efficient
1.2	83%	very efficient

Shao, X., Zhang, Z., Song, P., Feng, Y., & Wang, X. (2022). A review of energy efficiency evaluation metrics for data centers. Energy and Buildings, 112308.

Towards Sustainable Operation: Reuse Energy

- Traditional PUE does not account for energy reuse, like district heating, building heating, industrial heat recovery.

- **Energy Reuse Factor (ERF)** measures how much energy is reused outside the facility (waste heat).

- **Energy Reuse Effectiveness (ERE)** extends PUE by considering reused energy.
  - Lower ERE indicates more sustainable operation.

$$\begin{aligned}
\text{PUE} &= \frac{\text{total energy}}{\text{IT energy}} \\
&= \frac{\text{cooling} + \text{power} + \text{lighting} + \text{IT energy}}{\text{IT energy}}
\end{aligned}$$

$$\text{ERF} = \frac{\text{reuse energy}}{\text{total energy}}$$

$$\begin{aligned}
\text{ERE} &= \frac{\text{cooling} + \text{power} + \text{lighting} + \text{IT - reuse energy}}{\text{IT energy}} \\
&= (1 - \text{ERF}) \cdot \text{PUE}
\end{aligned}$$

Energy Efficiency Act in Germany (July 2026)

§ 11 Klimaneutrale Rechenzentren

(1) Rechenzentren, die vor dem 1. Juli 2026 den Betrieb aufnehmen oder aufgenommen haben, sind so zu errichten und zu betreiben, dass sie

1. ab dem 1. Juli 2027 eine Energieverbrauchseffektivität von kleiner oder gleich 1,5 und
2. ab dem 1. Juli 2030 eine Energieverbrauchseffektivität von kleiner oder gleich 1,3 im Jahresdurchschnitt dauerhaft erreichen.

(2) Rechenzentren, die ab dem 1. Juli 2026 den Betrieb aufnehmen, sind so zu errichten und zu betreiben, dass sie

1. eine Energieverbrauchseffektivität von kleiner oder gleich 1,2 erreichen und
2. einen Anteil an wiederverwendeter Energie nach DIN EN 50600-4-6, Ausgabe November 2020[7] von mindestens 10 Prozent aufweisen; Rechenzentren, die ab dem 1. Juli 2027 den Betrieb aufnehmen, müssen einen geplanten Anteil an wiederverwendeter Energie von mindestens 15 Prozent aufweisen; Rechenzentren, die ab dem 1. Juli 2028 den Betrieb aufnehmen, müssen einen geplanten Anteil an wiederverwendeter Energie von mindestens 20 Prozent aufweisen.

- https://www.gesetze-im-internet.de/enefg/BJNR1350B0023.html

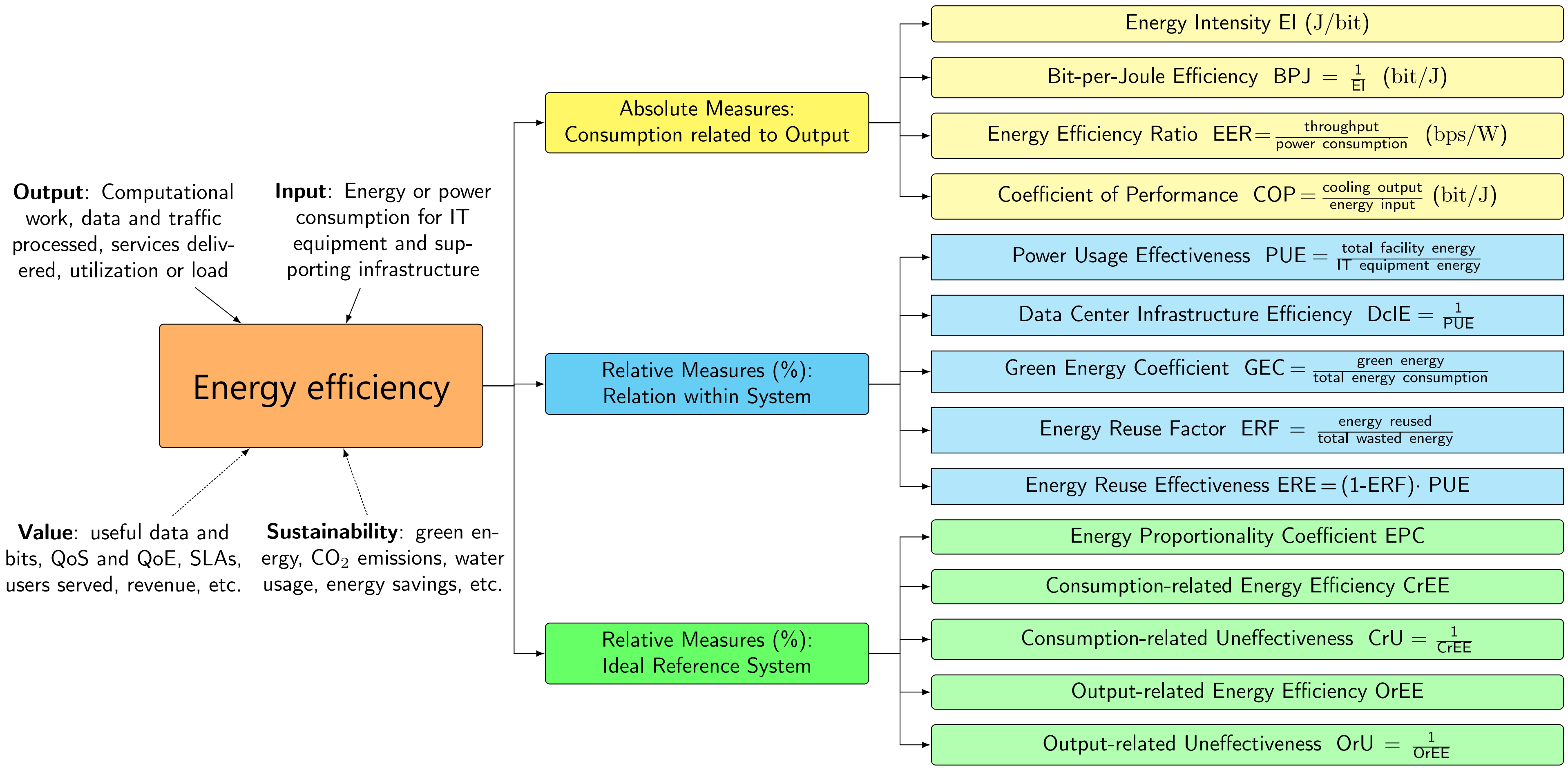
***Energieverbrauchseffektivität**: Verhältnis des jährlichen Energiebedarfs des gesamten Rechenzentrums zum Energiebedarf der Informationstechnik*

Power Usage Effectiveness PUE

Energy Reuse Factor (ERF)

Relative EE Measures: (Ideal) Reference System

Classification of EE Metrics: Reference Systems



Hossfeld, T., Wunderer, S., Loh, F., & Schien, D. (2024). Analysis of Energy Intensity and Generic Energy Efficiency Metrics in Communication Networks: Limits, Practical Applications and Case Studies. IEEE Access. https://doi.org/10.1109/ACCESS.2024.3435716

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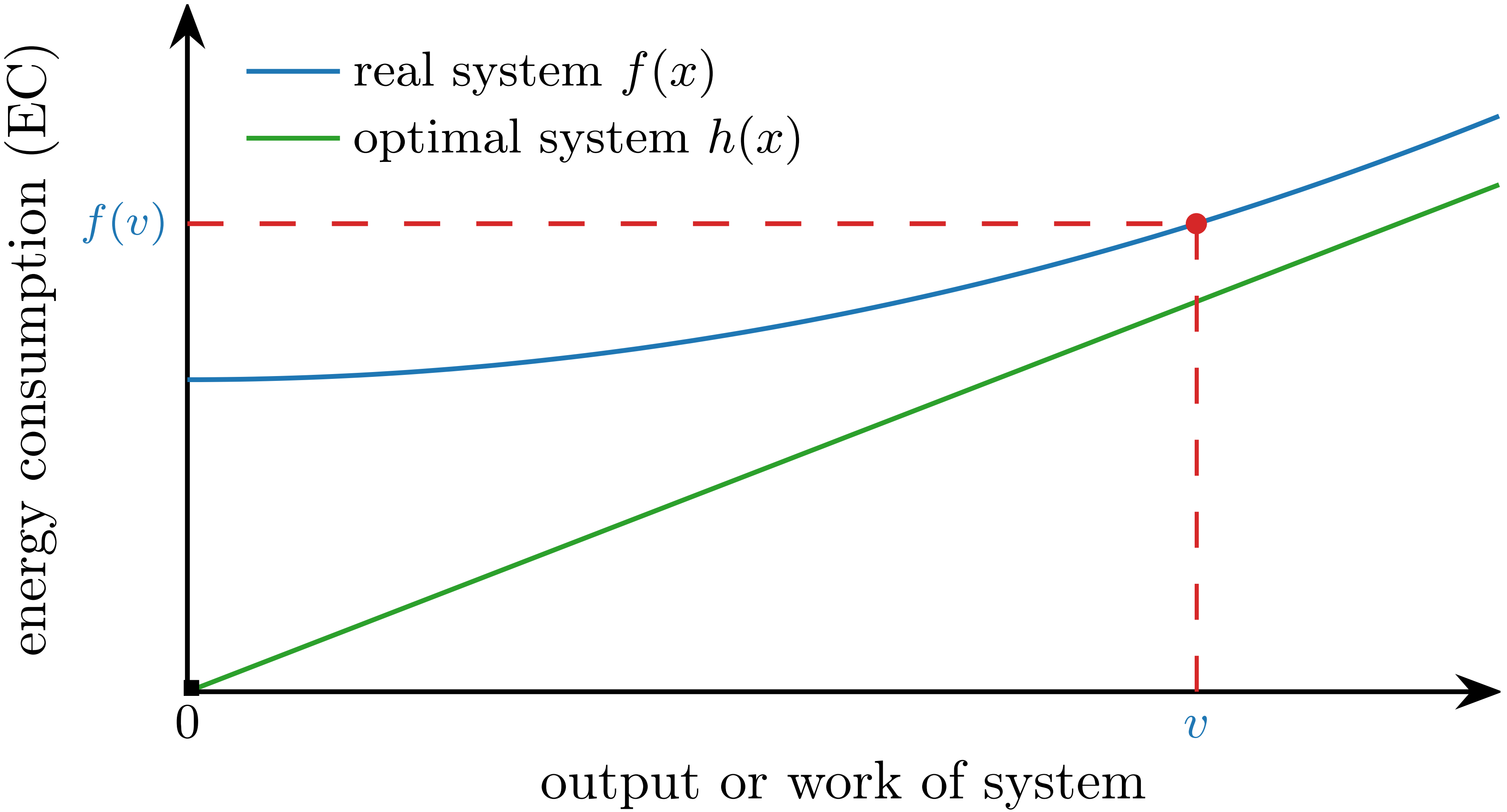
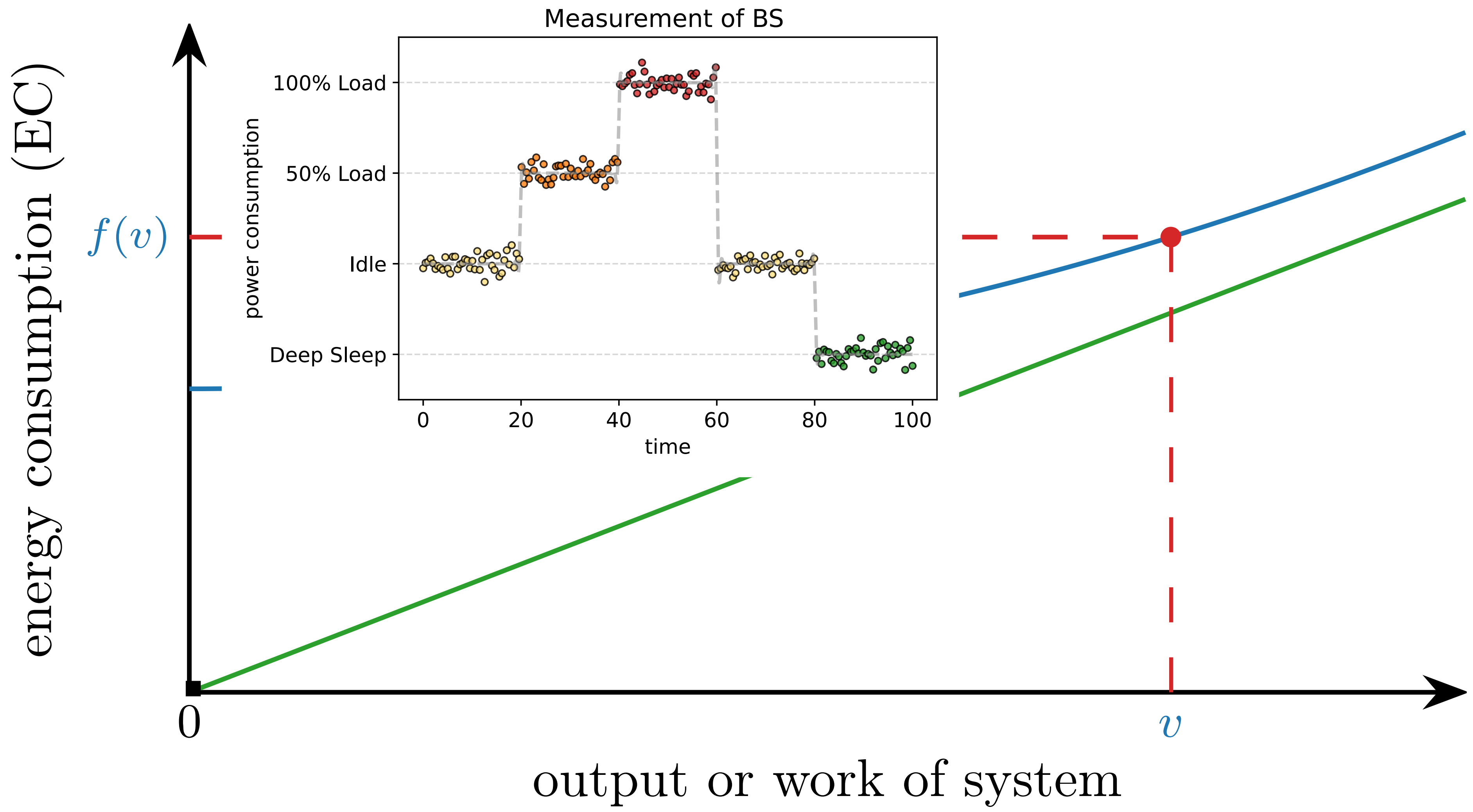
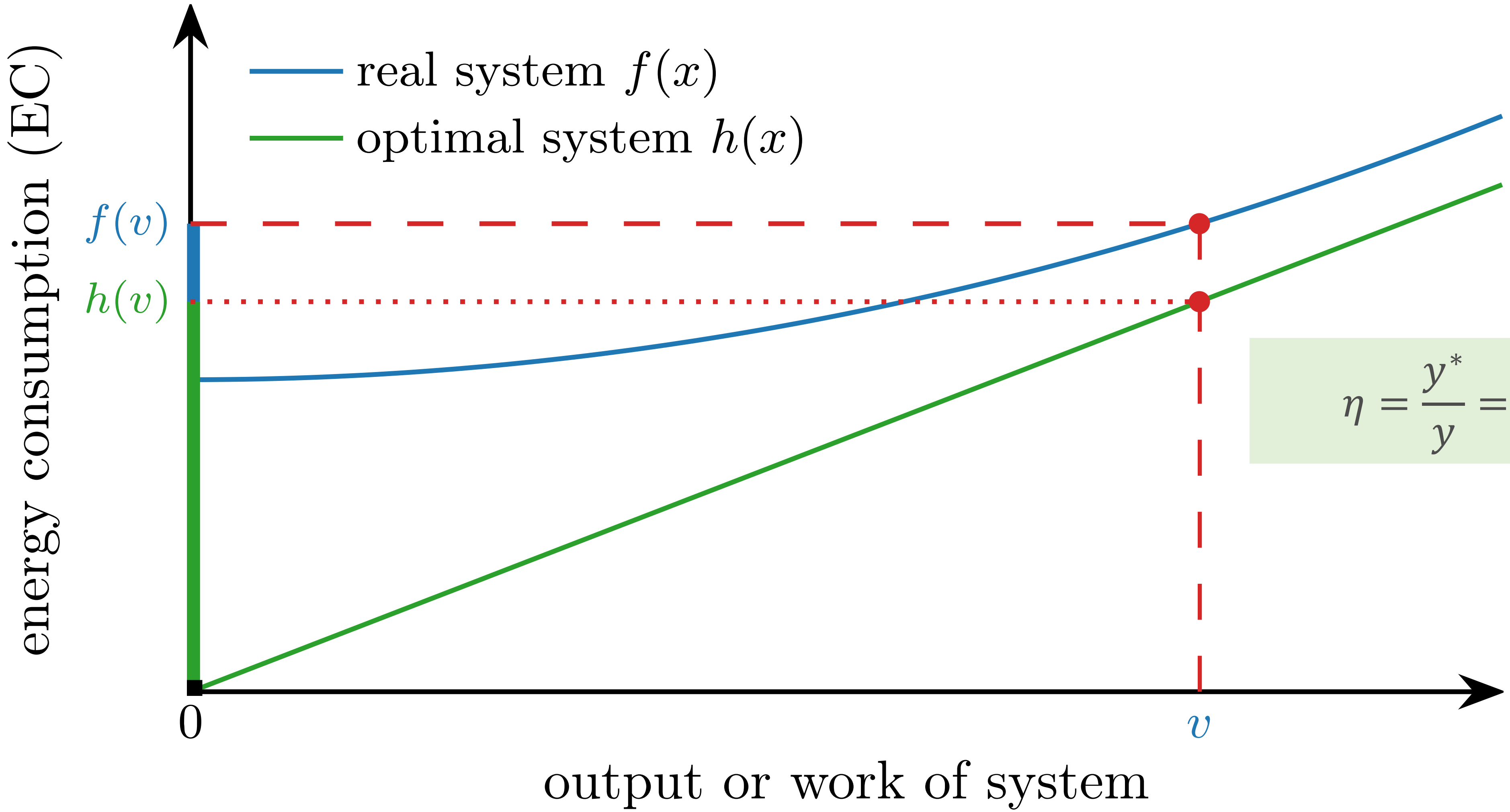
Visualization: Energy Efficiency as Relation to Optimal System



Visualization: Energy Efficiency as Relation to Optimal System

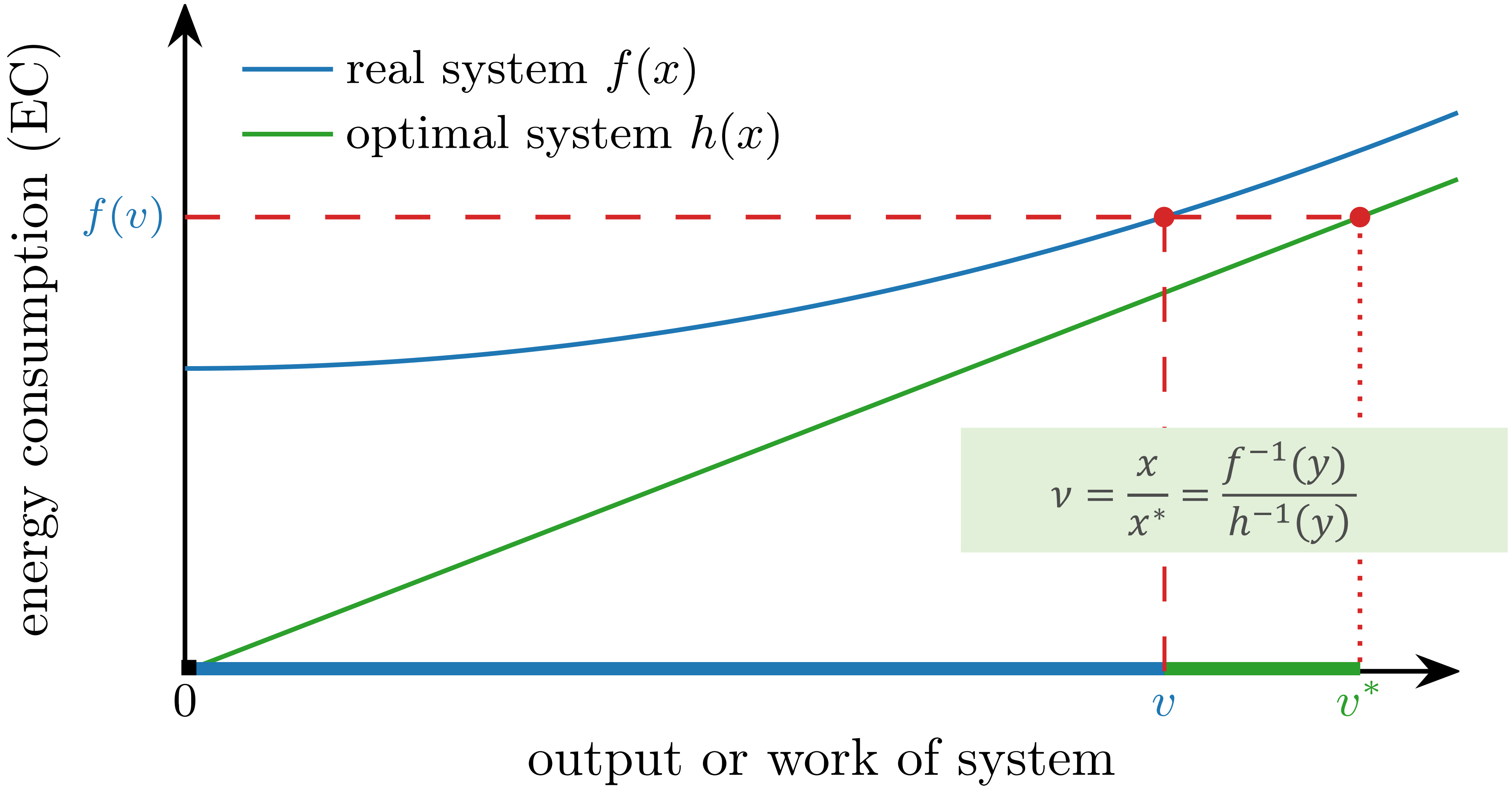


Visualization: Energy Efficiency as Relation to Optimal System (1)



$$\eta = \frac{y^*}{y} = \frac{h(x)}{f(x)}$$

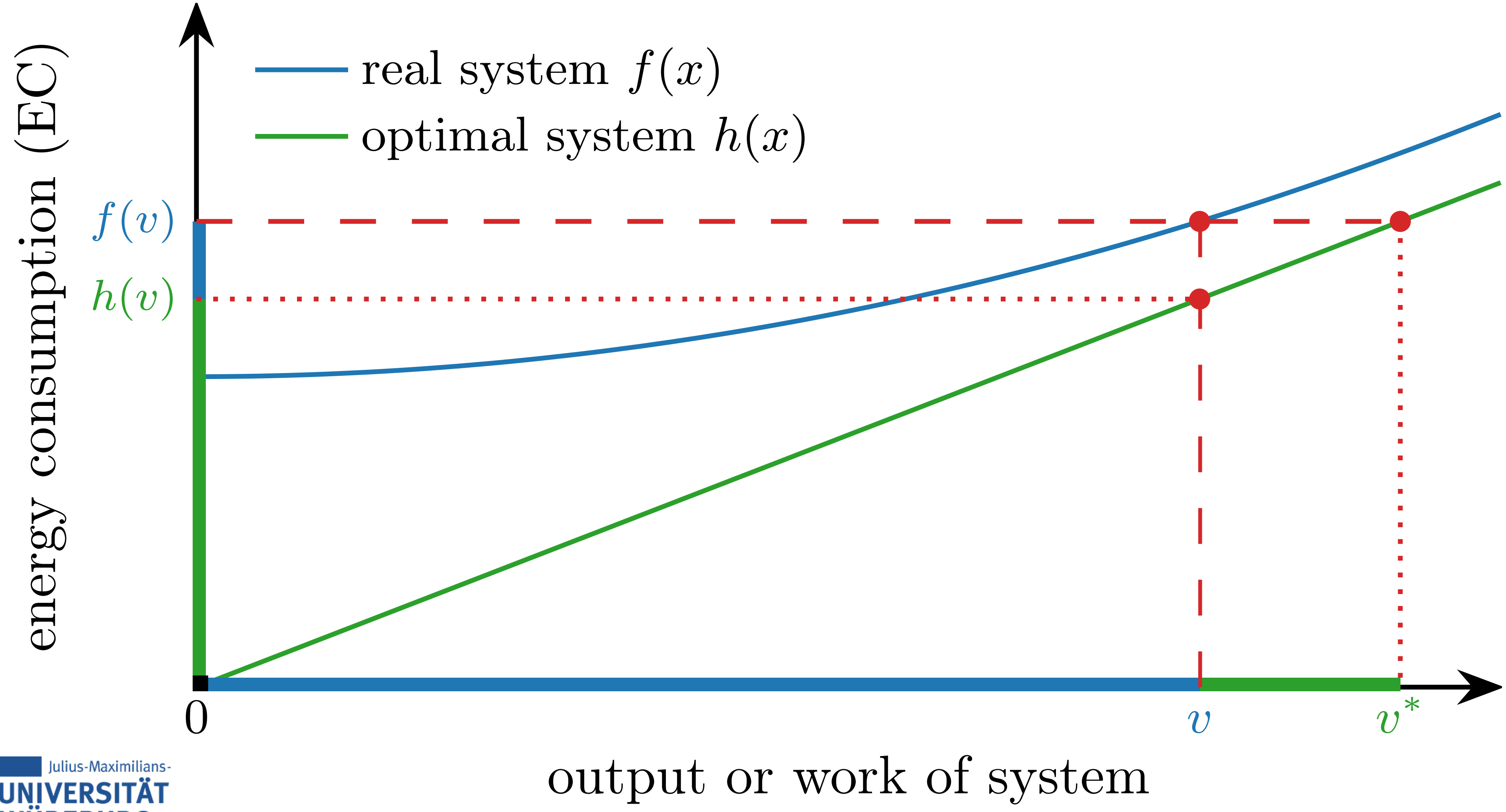
Visualization: Energy Efficiency as Relation to Optimal System (2)



Visualization: Energy Efficiency as Relation to Optimal System (3)

$$\eta = \frac{y^*}{y} = \frac{h(x)}{f(x)}$$

Consumption-related Energy Efficiency



Output-related Energy Efficiency

$$\nu = \frac{x}{x^*} = \frac{f^{-1}(y)}{h^{-1}(y)}$$

Energy Efficiency as Relation to (Ideal) Reference System

- Consider a **real system** (or part of a system)
  - measured energy consumption E
  - for processing the work/traffic v

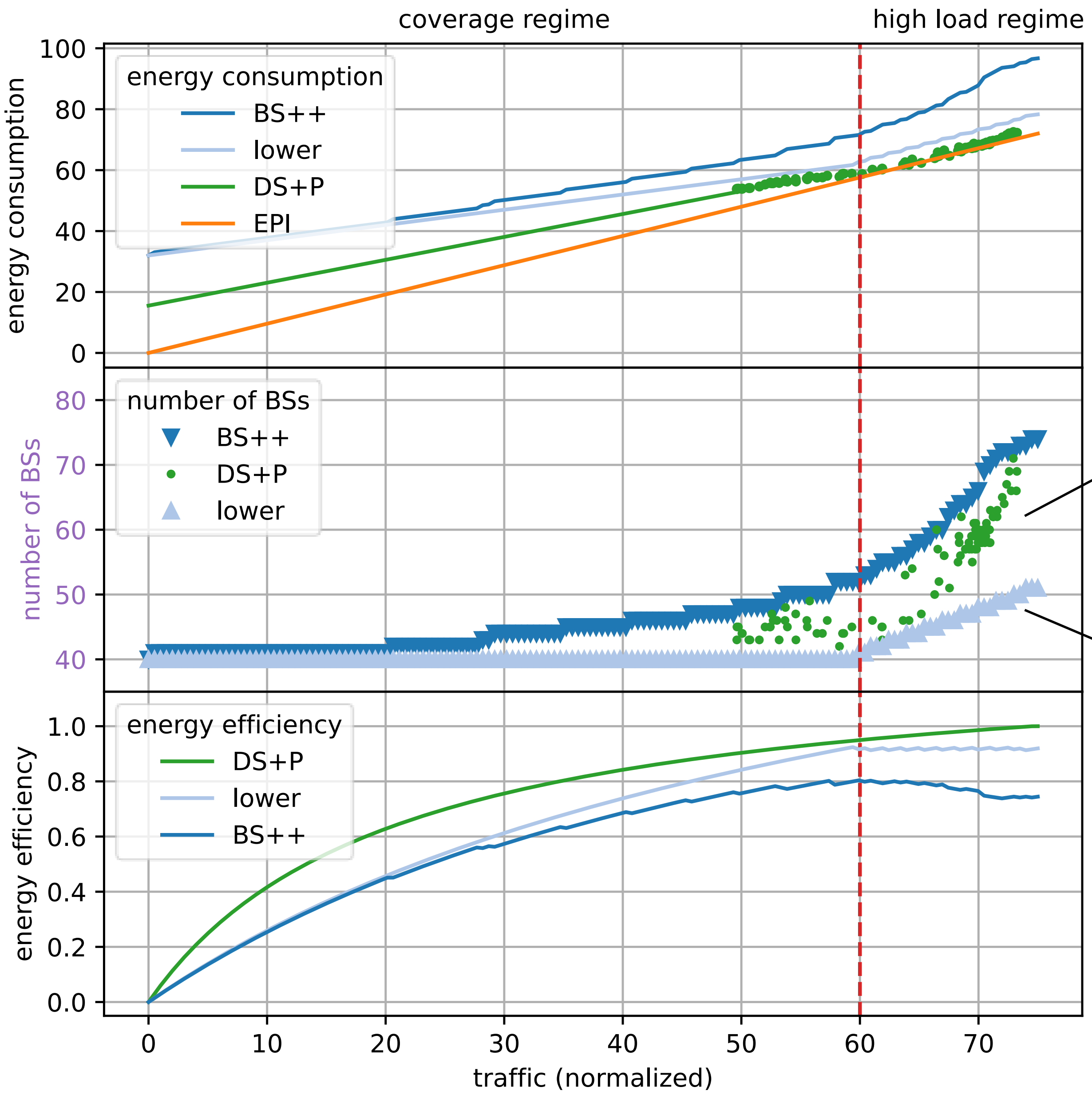
- Consider a **theoretically optimal, ideal system**
  - for processing the work/traffic v of the real system
  - the required energy consumption E^* of the ideal system is then $E^* \leq E$

$$\eta = \frac{E^*}{E}$$ **energy consumption related energy efficiency**

- Alternatively: consider a **theoretically optimal, ideal system**
  - With the energy consumption E, as measured in the real system,
  - the amount of work/traffic v^* that can be processed is then $v^* \geq v$

$$\nu = \frac{x}{x^*}$$ **output related energy efficiency**

Example: Results for 5G Networks



Deep sleep mode and spatio-temporal traffic prediction mechanism

Minimum number regarding coverage and traffic demand

Example: Optimal Energy Consumption for Modular Date Center Switches

Modular Linecard Power Management

- Modern high capacity routers e.g. Cisco ASR9000 or Juniper MX10008, are able to energize and de-energize subsystems and Interface Cards
  - Modern router linecards are built from identical modular *slices*, each containing several physical interfaces.
  - Each slice can independently enter a low-power state.
  - If all ports of a slice are unused:
    - the slice can be administratively disabled,
    - reducing linecard power consumption significantly.
  - Other slices continue operating normally.

Plugin part number	Gbit/sec	Watt
Arista SFP-10G-ZR	10	1
Arista QSFP-40G-ER4	40	3.5
Arista QSFP-ZR4-100G	100	5.5
Arista QDD-400G-LR	400	12
Arista QDD-800G-2LR4	800	16

Numbers taken from https://www.cisco.com

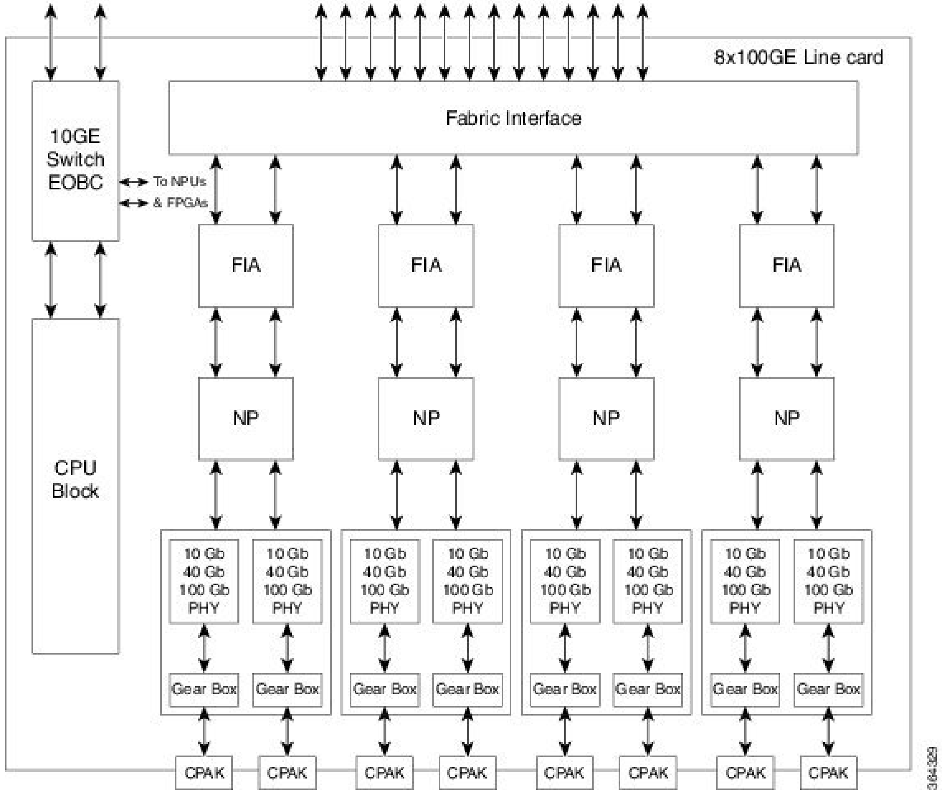
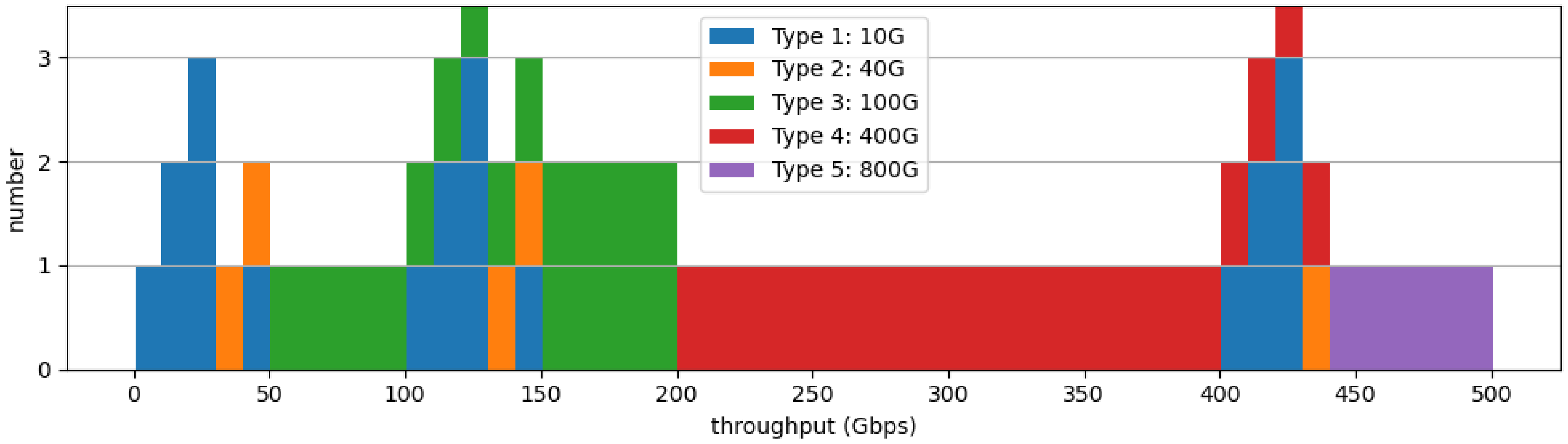


Figure taken from https://www.cisco.com

Optimization Problem: Minimal Number of Plugin Types

- Each type-j plugin contributes a fixed amount of throughput B_j and consumes power β_j.
- Goal: optimal combination of energized plugins of different types that satisfies a required total throughput bit rate, x, while minimizing total power.
  - Plugin counts m_j as non-negative integer decision variables
  - throughput data x rate as a constraint



$$min \sum_{j=1}^{N} m_j \beta_j$$

$$\text{such that} \sum_{j=1}^{N} m_j B_j \geq x \quad \text{with} \quad m_j \in \mathbb{N}_0 \,, \forall j \in \{1, \ldots, N\}$$

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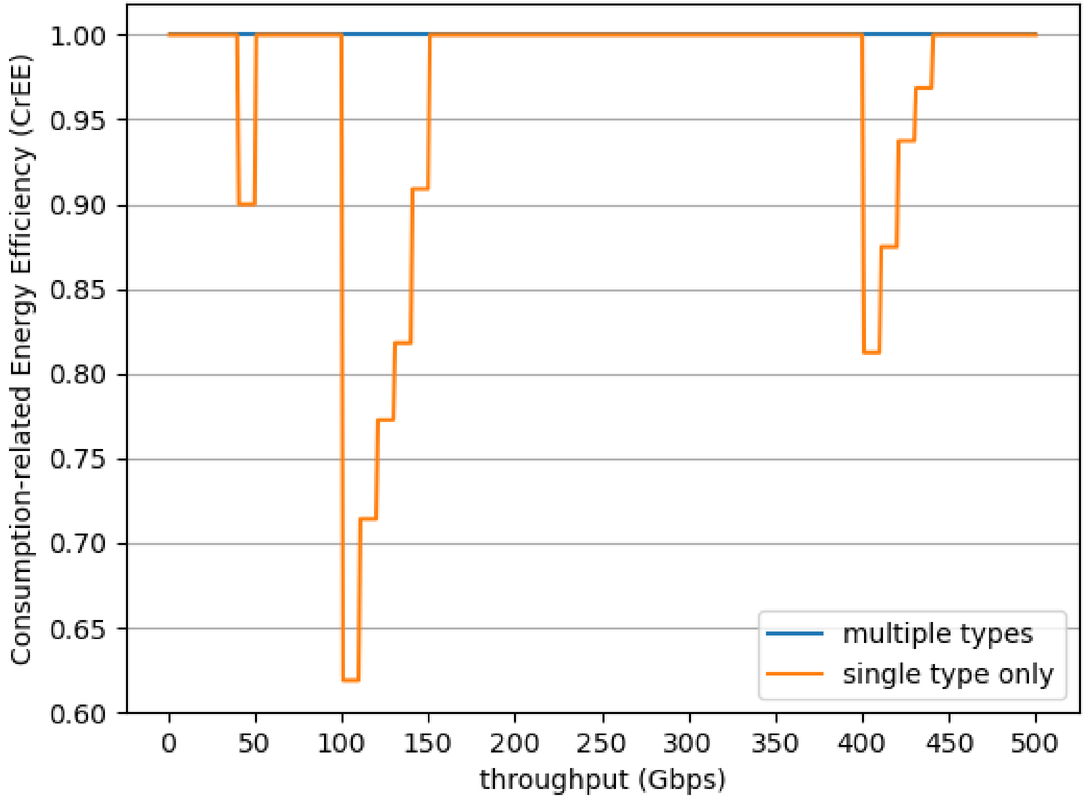
Consumption-related Energy Efficiency

- Compare the optimal solution using multiple types with the simplest solution
  - single plugin type which provides sufficient capacity for the required throughput

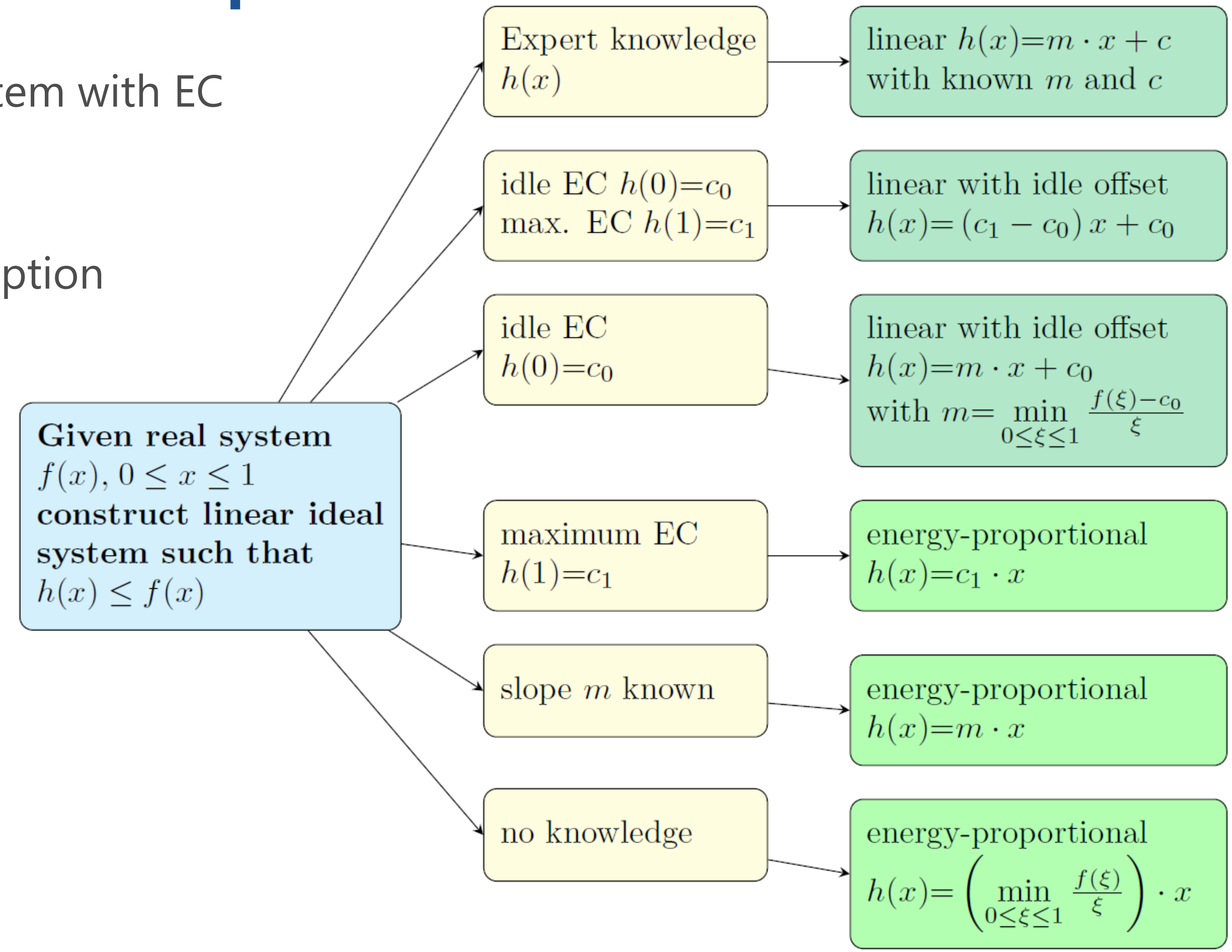
- CrEE indicates throughput ranges which may suffer from the simplest solution



Linear Reference Systems: “Top-Down”

- Ideal: energy-proportional system with EC
  $f(x) = mx$ for traffic x

- In practice: idle power consumption
  (e.g. to offer availability)
  $f(x) = mx + c$ with offset



Hoßfeld, T. (2025). A framework for generic energy efficiency metrics: Defining ideal linear reference systems. IEEE 2025 EuCNC/6G Summit.

Beyond Energy Efficiency

Efficiency with respect to Sustainability

Absolute measures

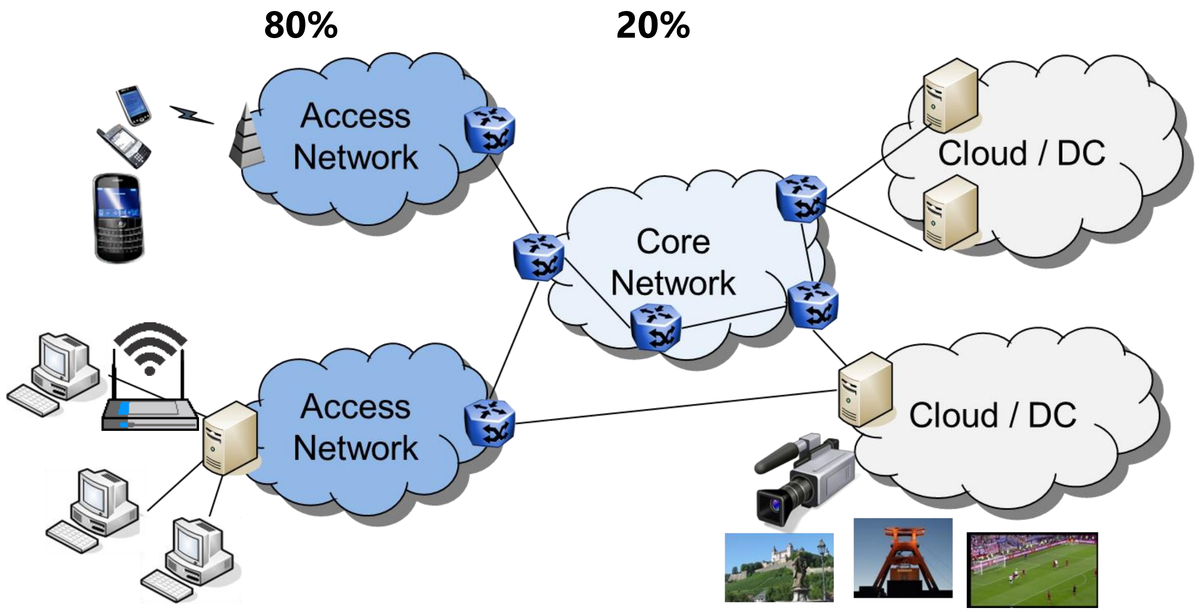
- Water usage effectiveness in data centers $\mathrm{WUE} = \frac{\text{amount of water for cooling equip.}}{\text{total energy consumed}}$ (L/J)
- Carbon emission factor $\mathrm{CEF} = \frac{\text{total green house gas emmisions}}{\text{total energy consumed}}$ (CO2e/J)
- Carbon usage effectiveness $\mathrm{CUE} = \frac{\text{total green house gas emmisions}}{\text{IT equip. energy consumption}}$ (CO2e/J)
- Carbon intensity $\mathrm{CI} = \frac{\text{total green house gas emmisions}}{\text{data traffic volume}}$ (CO2e/Byte)

Relation within the system

- Green energy coefficient $\mathrm{GEC} = \frac{\text{green energy}}{\text{total energy consumed}}$
- Energy Reuse Factor $\mathrm{ERF} = \frac{\text{energy reused}}{\text{total energy consumption}}$
- Energy Reuse Effectiveness $\mathrm{ERE} = (1 - \mathrm{ERF}) \cdot \mathrm{PUE}$
- Note PUE is in this category (dimensionless)

Energy consumption on the Internet

- By operating segment
  - Networking
    - Access Network
    - Core Network
  - Device
    - Laptop, PC, SmartTV, Smartphone, …
    - WLAN access
  - Cloud / Data Center
    - Storage of Data
    - Processing of Data



- Manufacturing and Production
  - Devices
  - Content such as videos, etc.
  - Learning of AI and ML models

in particular, CO_2 emissions (Life Cycle Assessment)

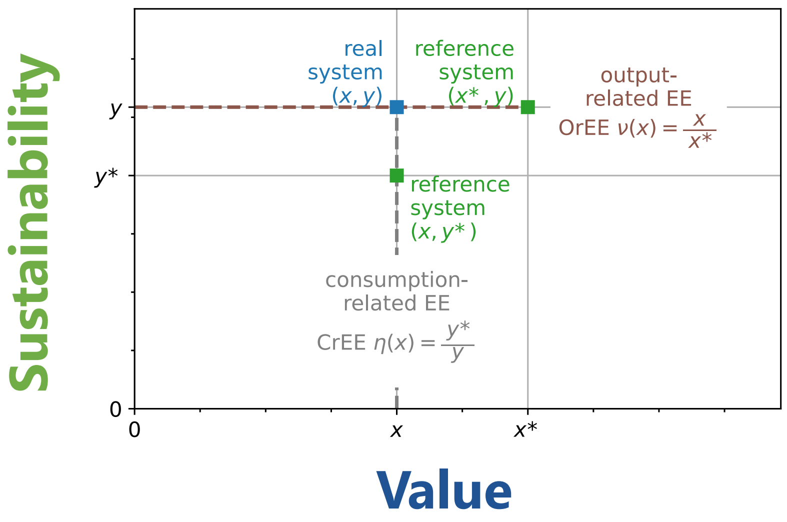
Energy Efficiency Framework

Input: energy or power consumption for IT equipment, servers, networking, and supporting infrastructure

Output: computational work, networking, data and traffic processed, services delivered, utilization, load

Energy efficiency
(of system, subsystem, device, component)

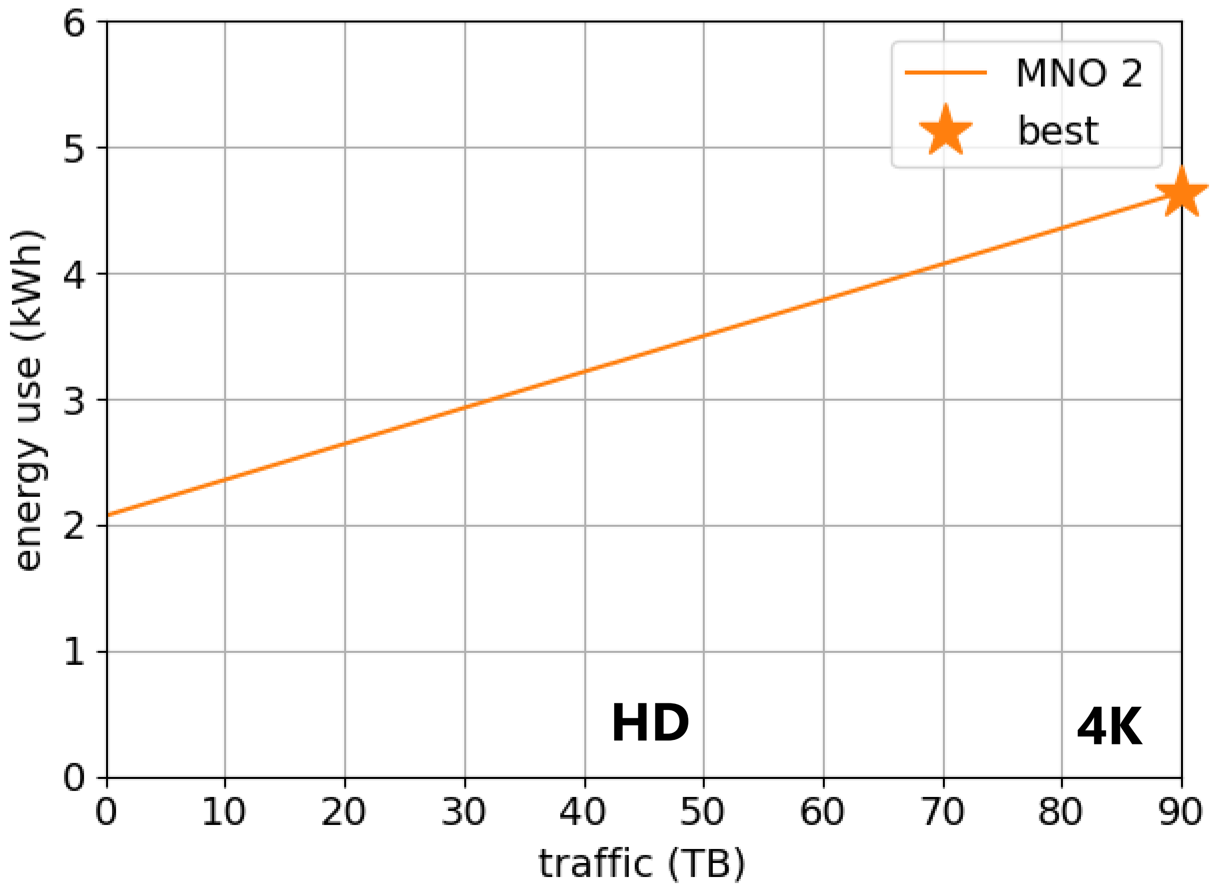
Sustainability: emissions, green energy usage, water usage, …



Value: useful bits, fulfillment of SLAs, QoE, KVIs, …

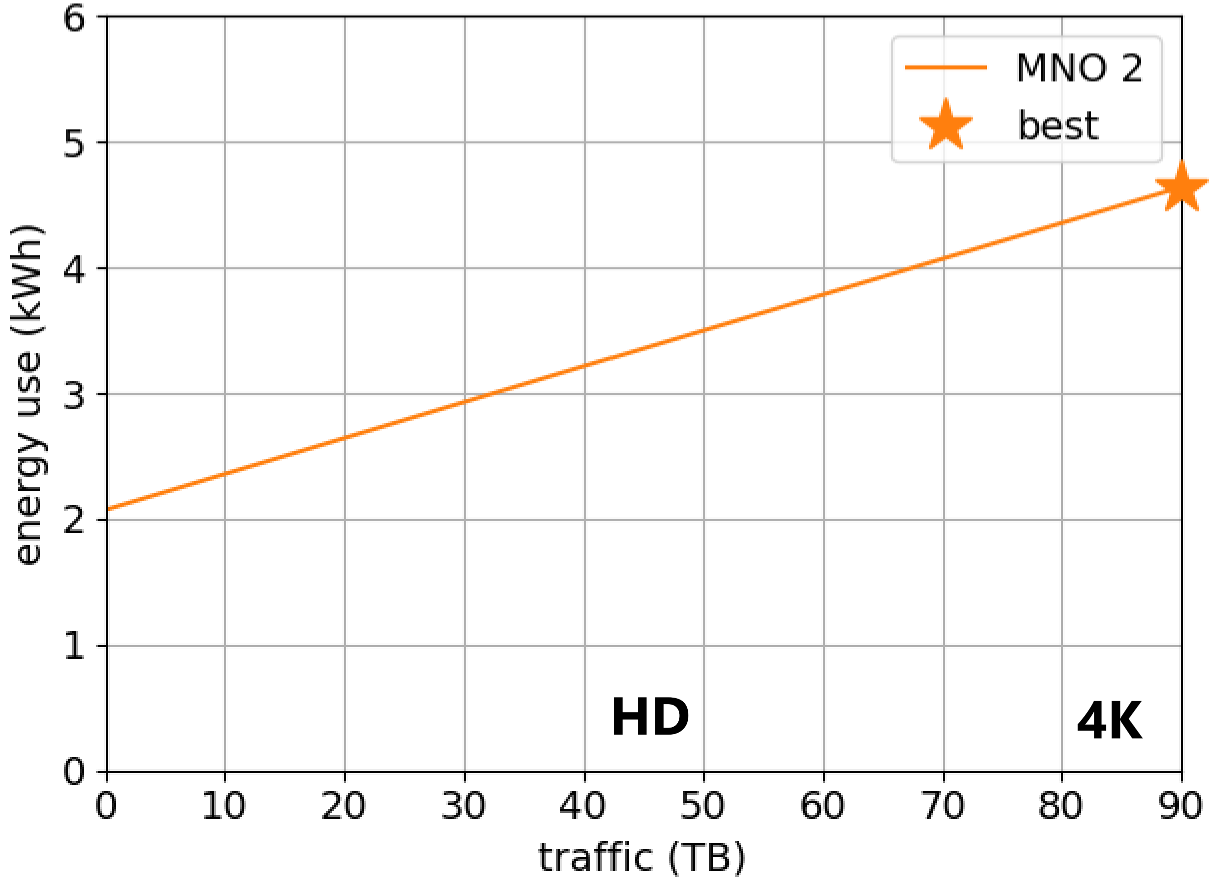
Quality of Experience (QoE) instead of Bits

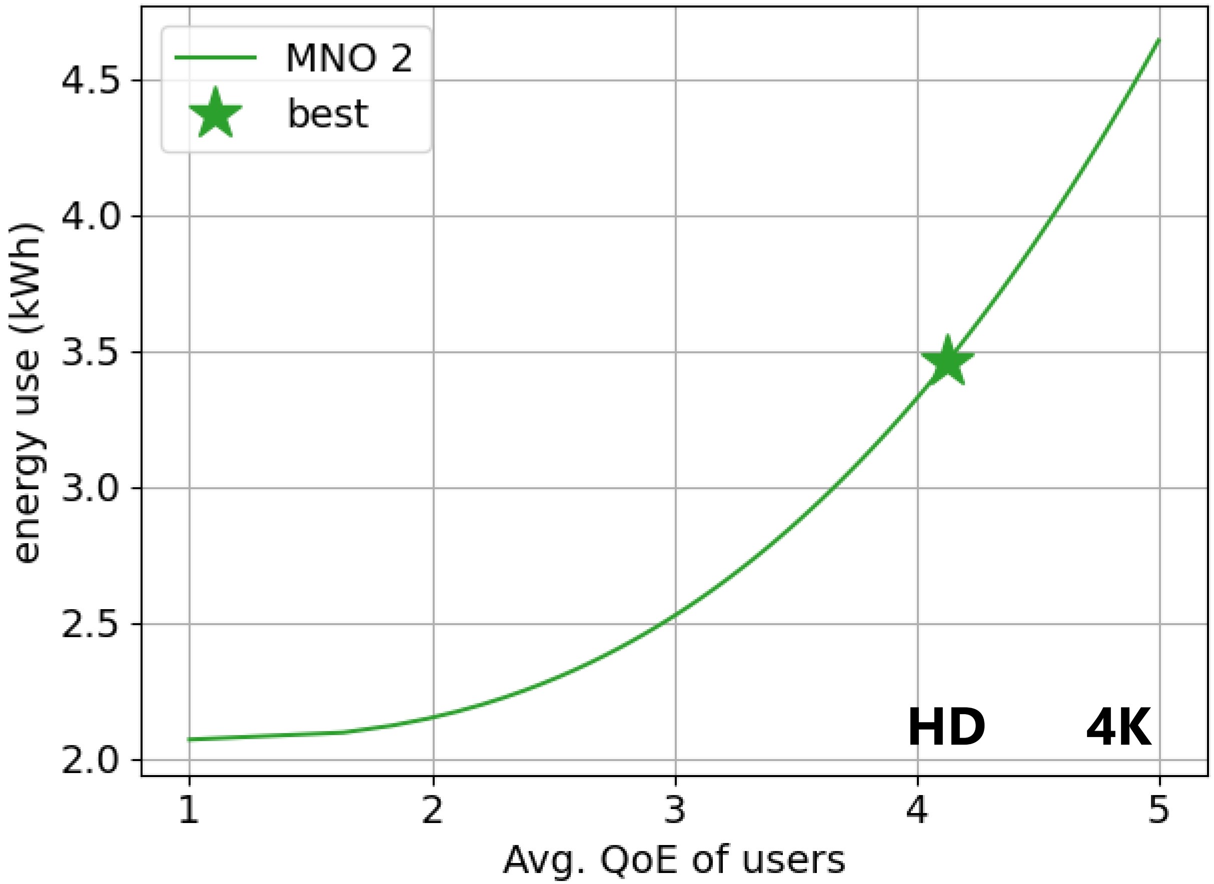
- Relate the service quality (QoE, SLAs, …) to the actual costs wrt. energy use
- The metric will change the decisions how to configure / improve networks



Quality of Experience (QoE) instead of Bits

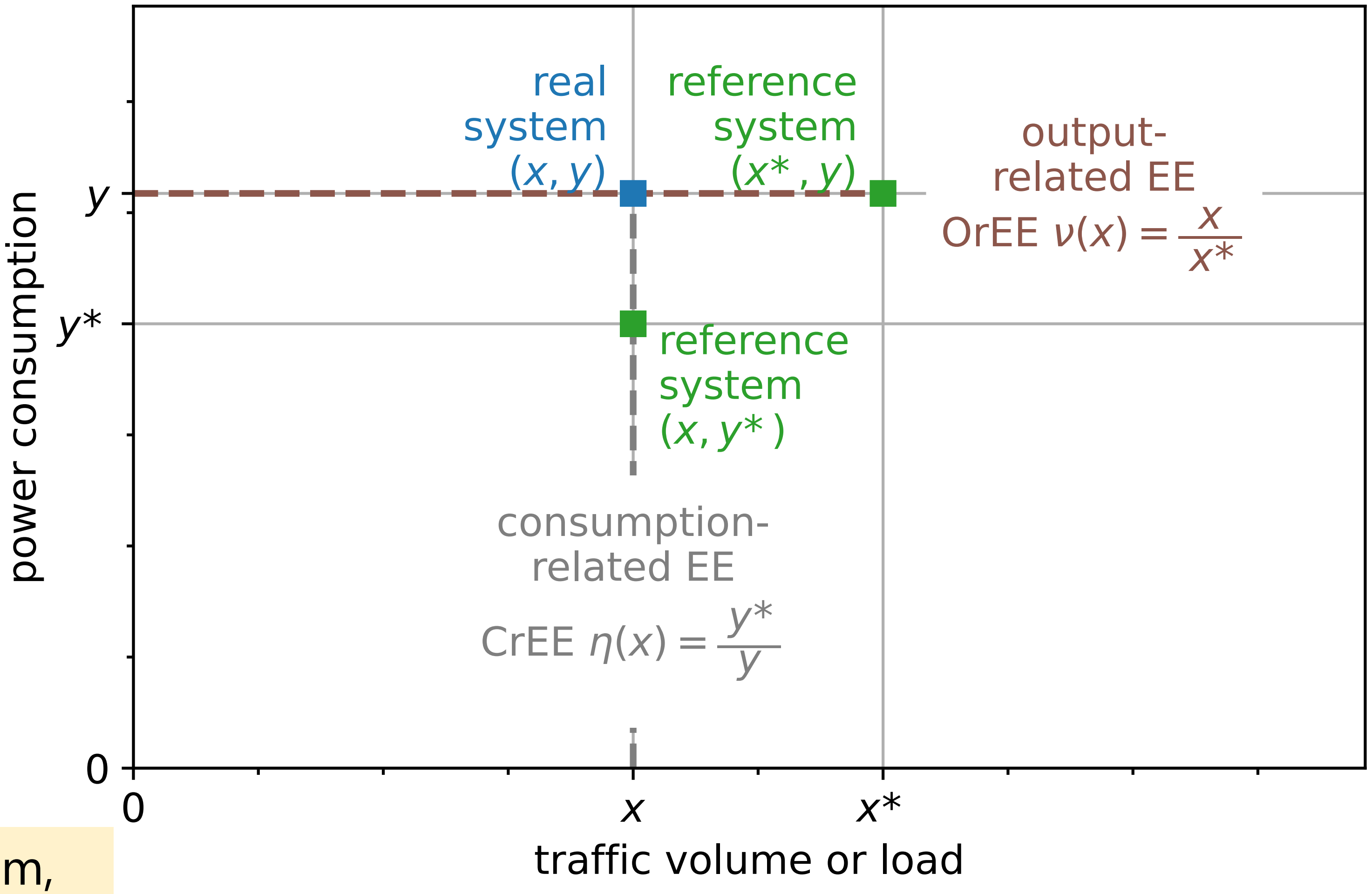
- Relate the service quality (QoS, QoE, SLAs, …) of users to the actual costs wrt. energy use
- The metric will change the decisions how to configure / improve networks





Hossfeld, T., Varela, M., Skorin-Kapov, L., & Heegaard, P. E. (2023). A Greener Experience: Trade-offs between QoE and CO2 Emissions in Today's and 6G Networks. IEEE Communications Magazine.

Framework: Definition of Metrics and Reference System



Sustainability: emissions, green energy usage, water usage, …

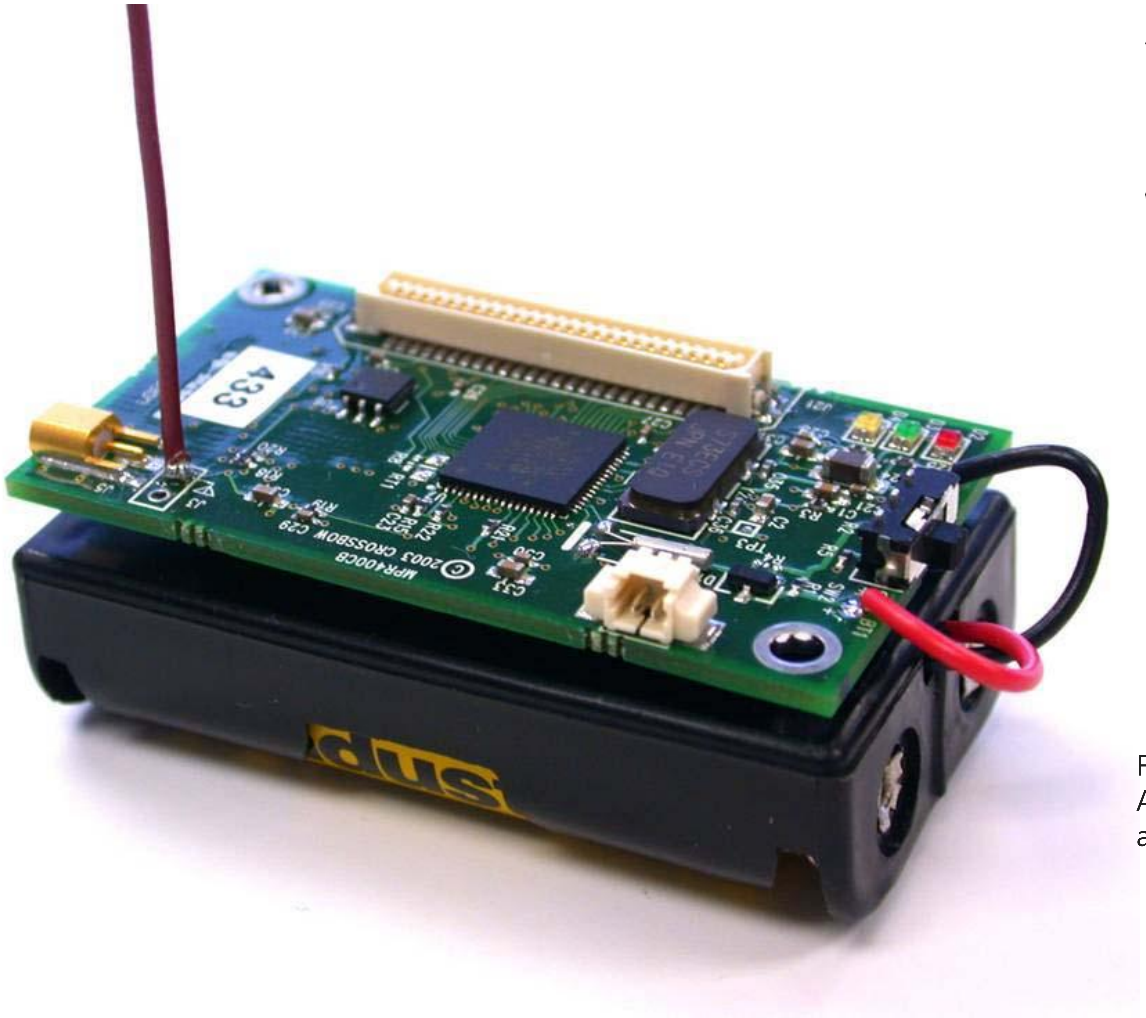
Reference: target system, comparison between operators, …

Value: useful bits, fulfillment of SLAs, QoE, KVIs, …

Markov Reward Model

Just for illustration

Example: Sensor node



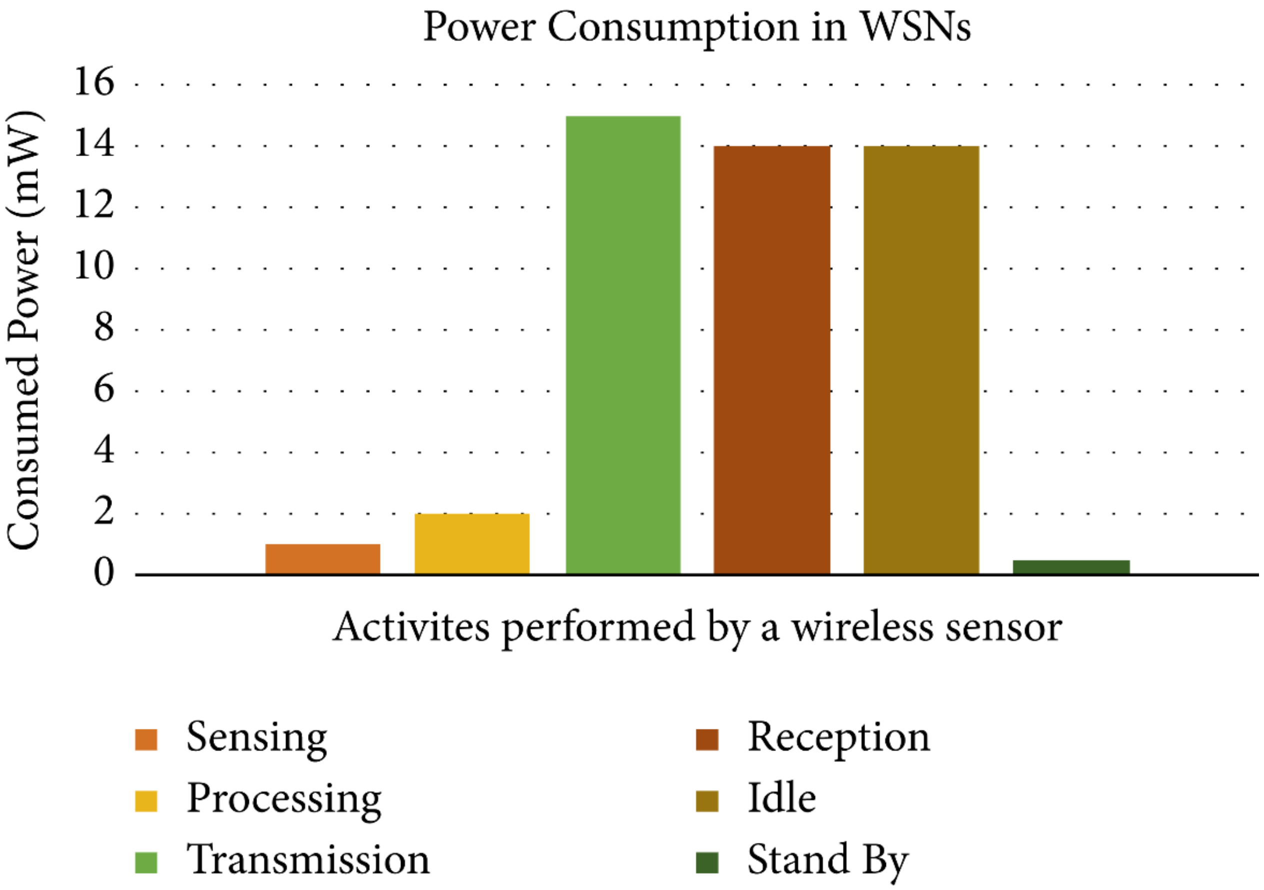
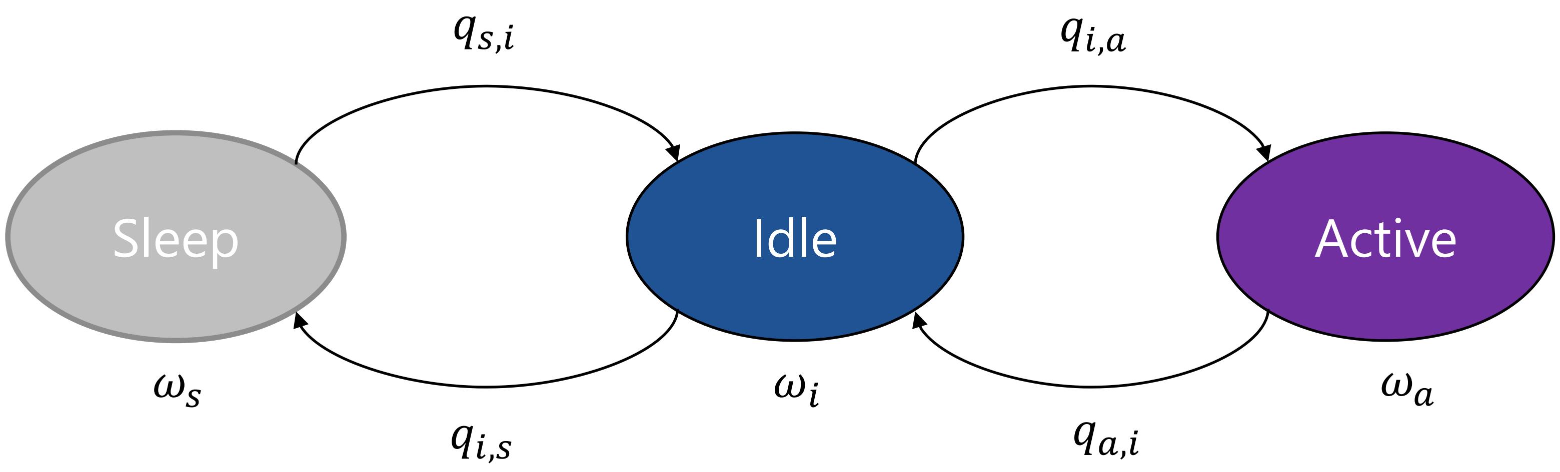


Figure taken from
Amengu, Angelina Ankah, et al. "SMAC-Based WSN Protocol-Current State of the Art, Challenges, and Future Directions." *Journal of Computer Networks and Communications* 2022 (2022).

Markov Reward Model

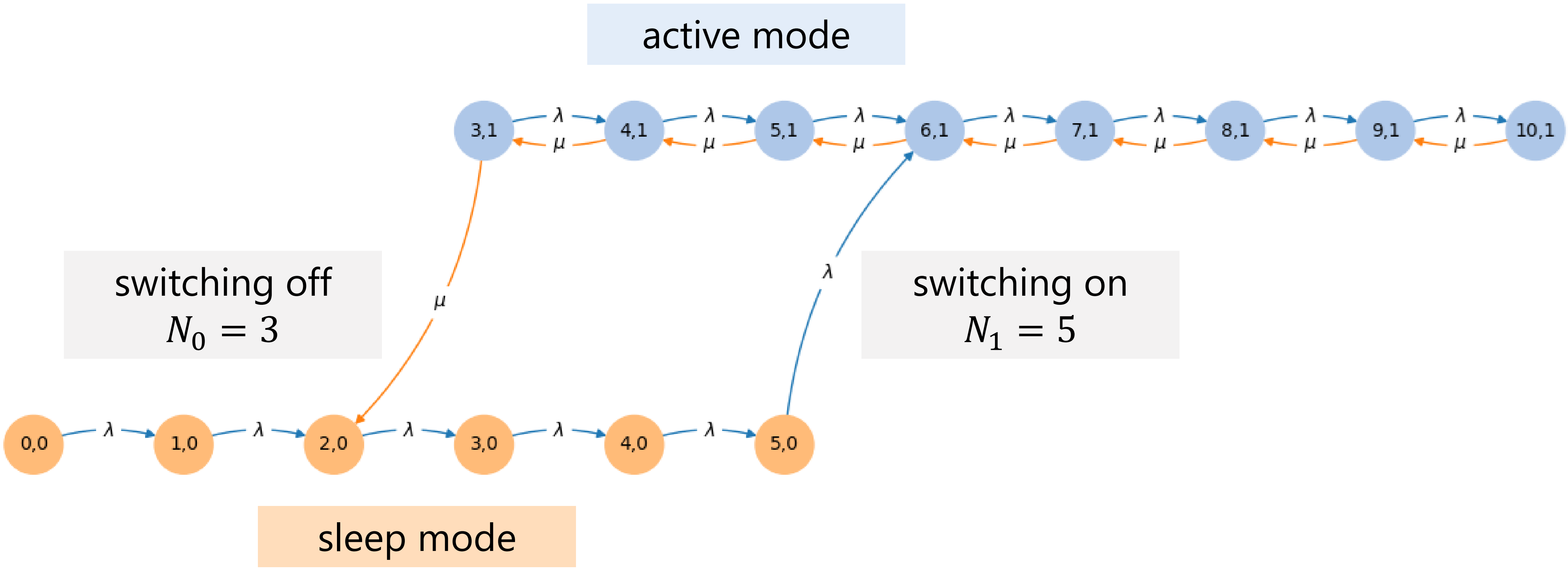
- Markov model
  - rate q_{ij} between states i and j captures the frequency of transitions
  - yields steady state probability $P(X = i) = x(i)$ to be in state i by by solving $X \cdot Q = 0$ (global balance equation)
- **Reward** per state: average power consumption ω_i
- **Expected power consumption**

$$\Omega = \sum_i \omega_i \cdot P(X = i)$$

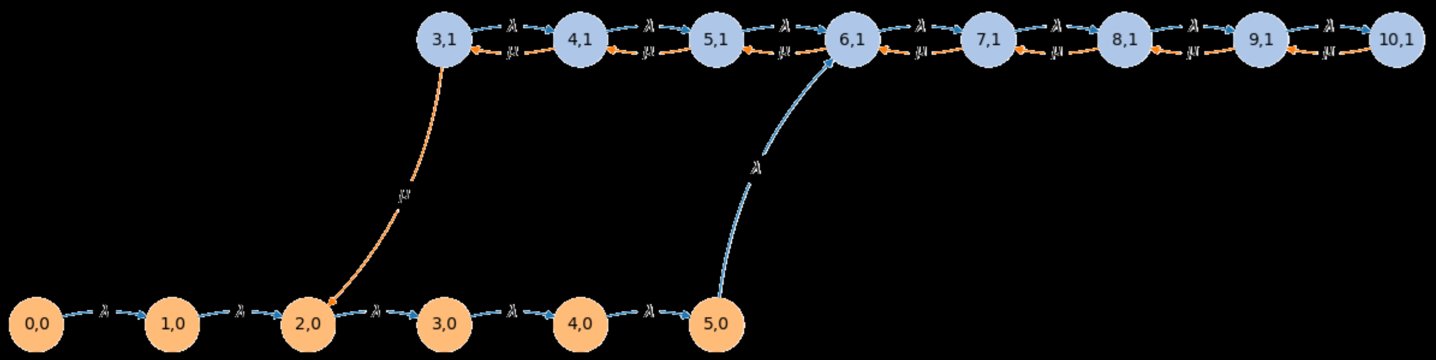


Switching on: Hysteresis Thresholds

- State indicates (X, Y) with number X of active jobs and current number of active servers Y

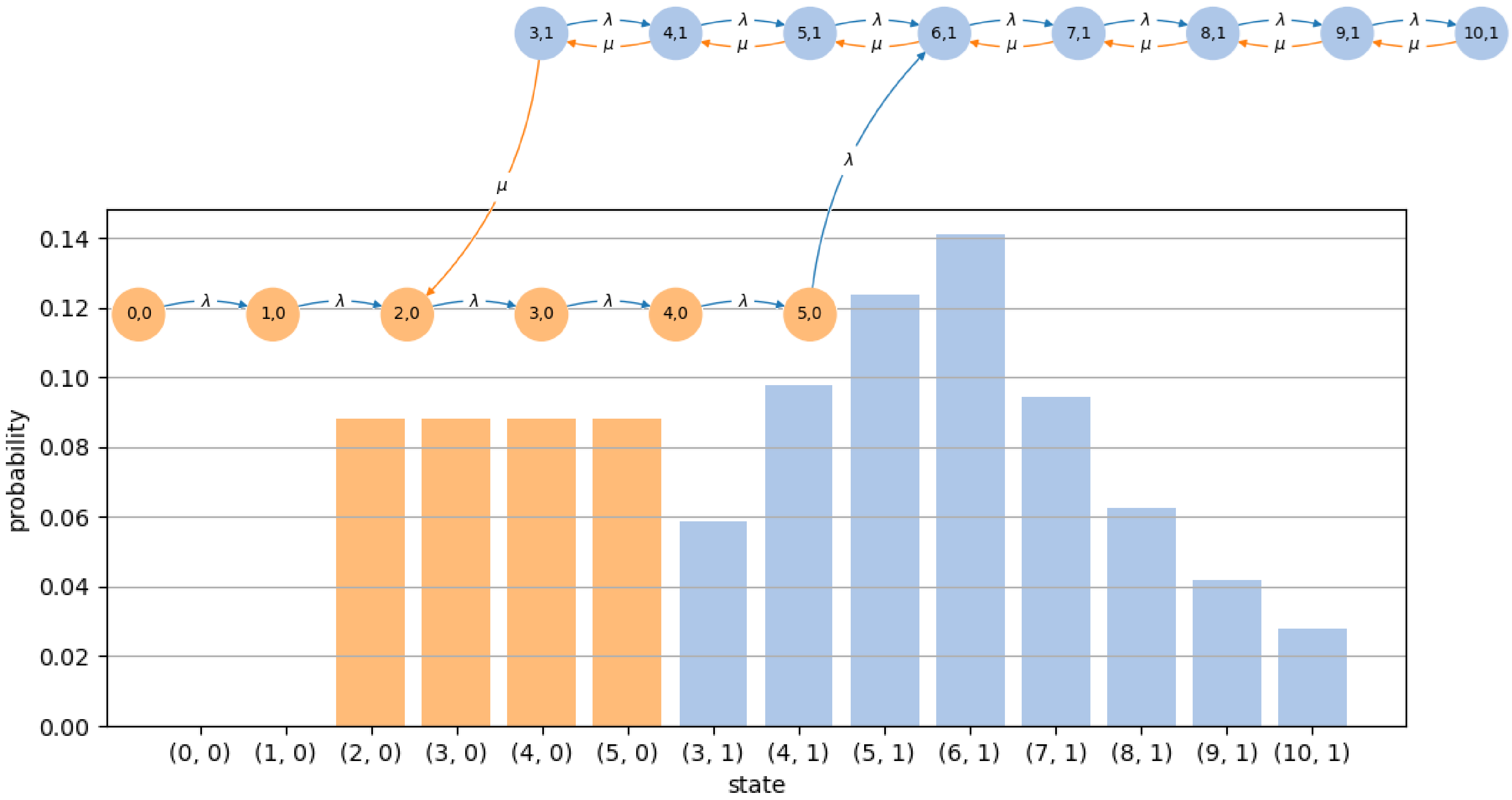


```
G = createMarkovModel() # transition graph
G.drawTransitionGraph()
probs = G.calcSteadyStateProbs()
```



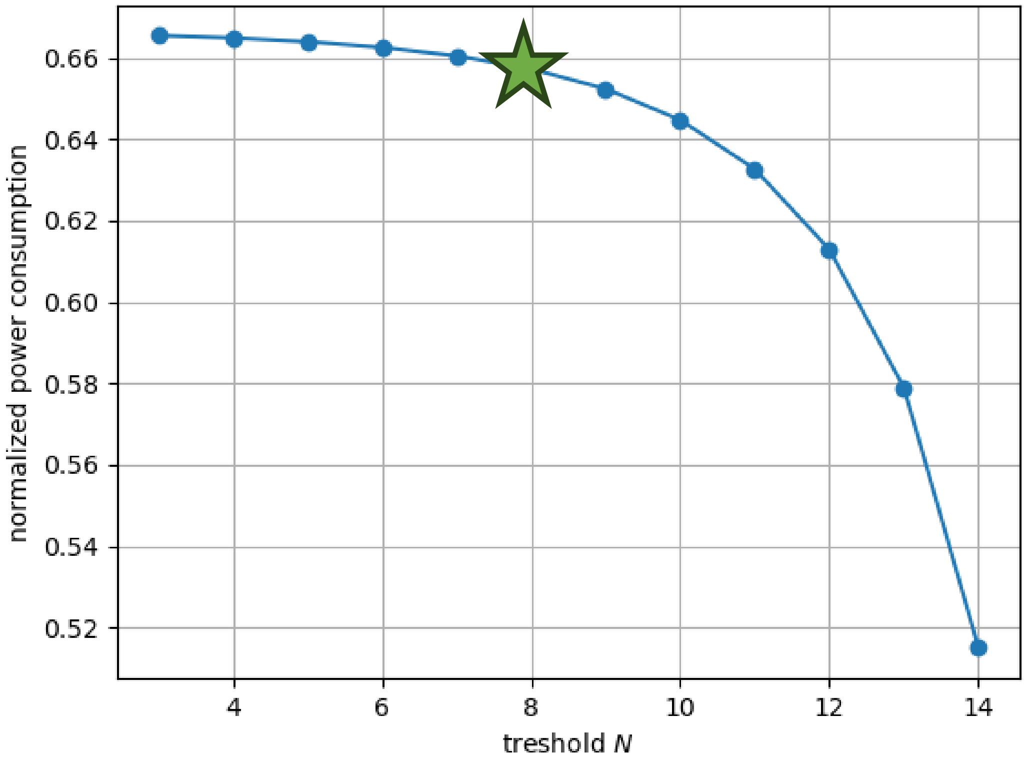
https://www.performance-modeling.org/scripts/

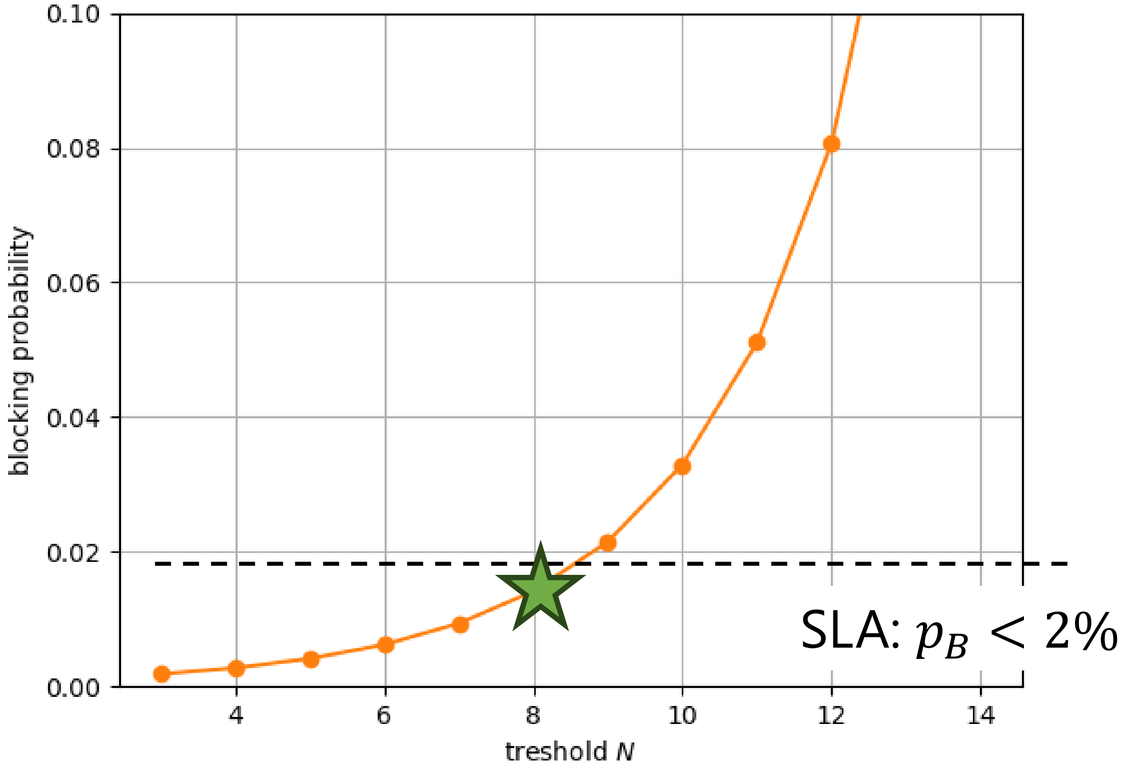
Example Result: Steady State Probabilities



Parameter Study

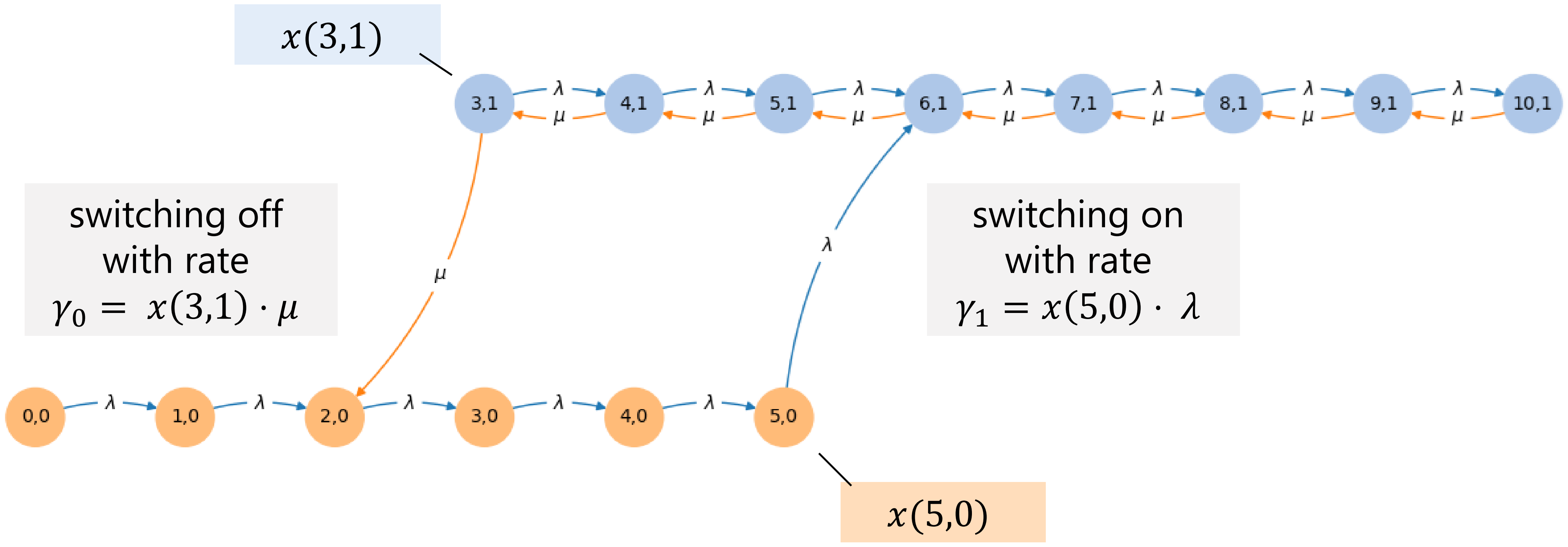
- Threshold N indicates when activating the server
- Server is only switched off if empty





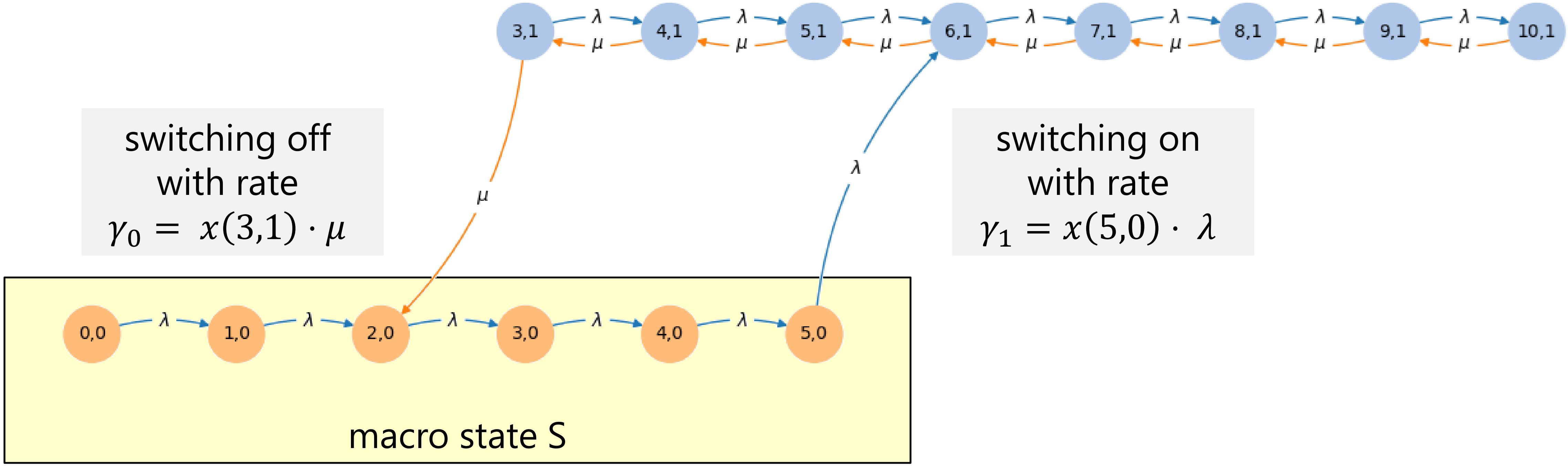
Rate of Device Power-On/Off Events

- Probability of being in that state x the transititon rate



Rate of Device Power-On/Off Events

- Probability of being in that state x the transitition rate



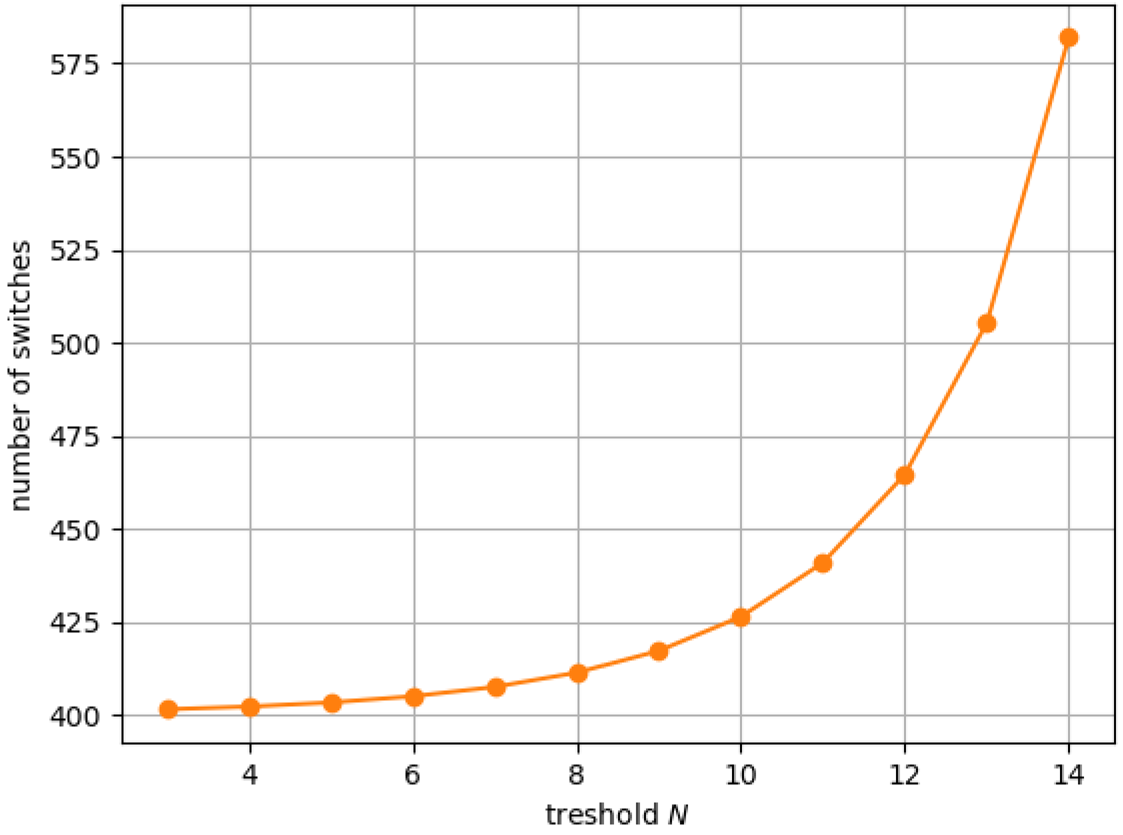
switching off with rate $\gamma_0 = x(3{,}1) \cdot \mu$

switching on with rate $\gamma_1 = x(5{,}0) \cdot \lambda$

balance equation for macro state S: $\gamma_1 = \gamma_0 \coloneqq \gamma$

Number of Device Power-On Events

- During a fixed time interval T, the number of events is $k \approx \gamma \cdot T$



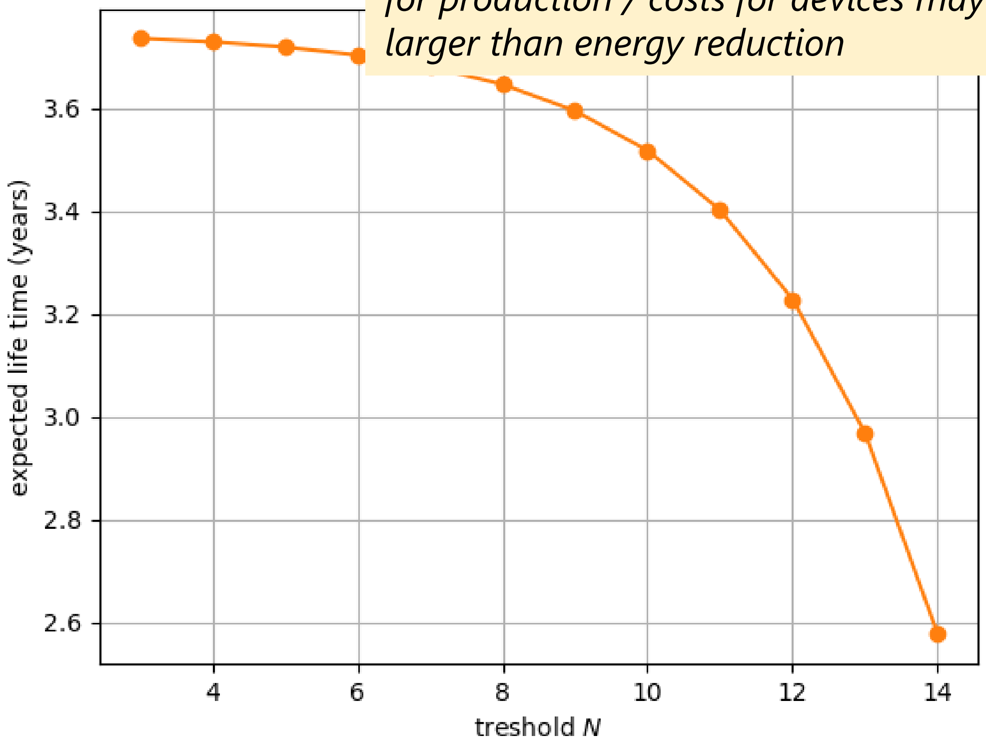
The LED lamp is rated for 20,000 operating hours and 15,000 switch cycles.

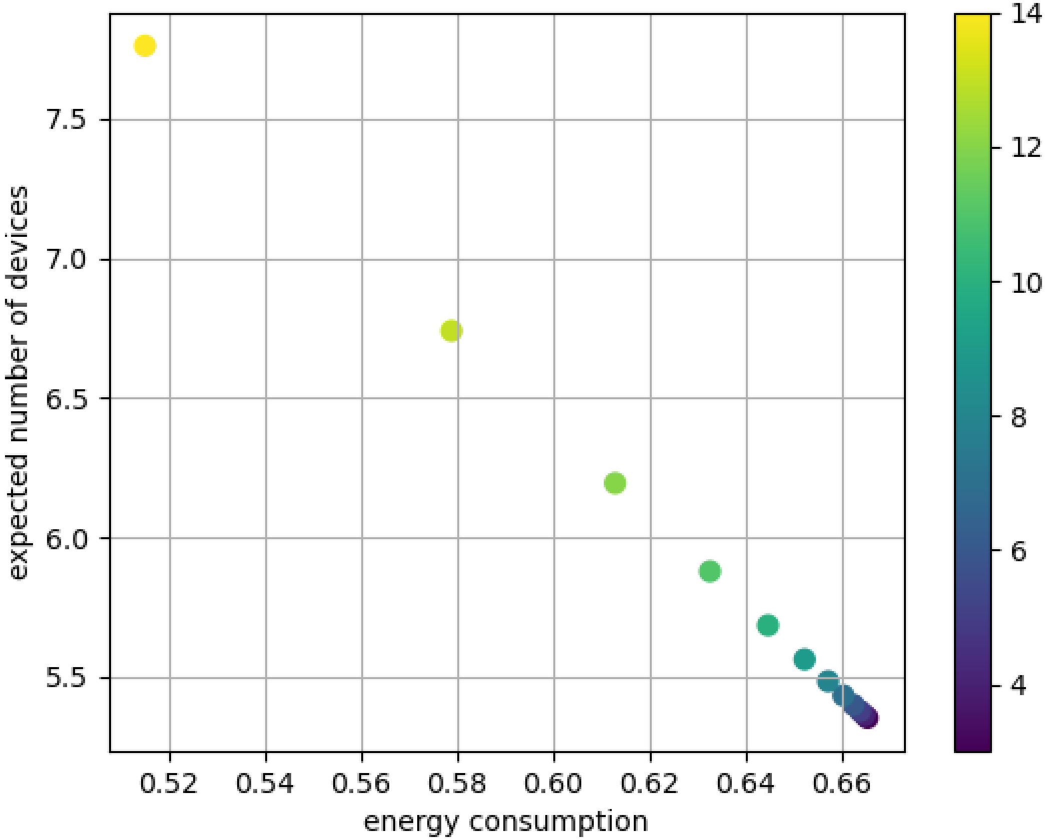
Lifetime decreases with increasing number of on/off cycles.

Lifetime of Device

- During a fixed time interval T, the number of events is $k \approx \gamma \cdot T$
- Mechanism saves energy … … but increases embodied carbon emissions !

But without switching mode, the lifetime would be 15 years … embodied emissions for production / costs for devices may be larger than energy reduction

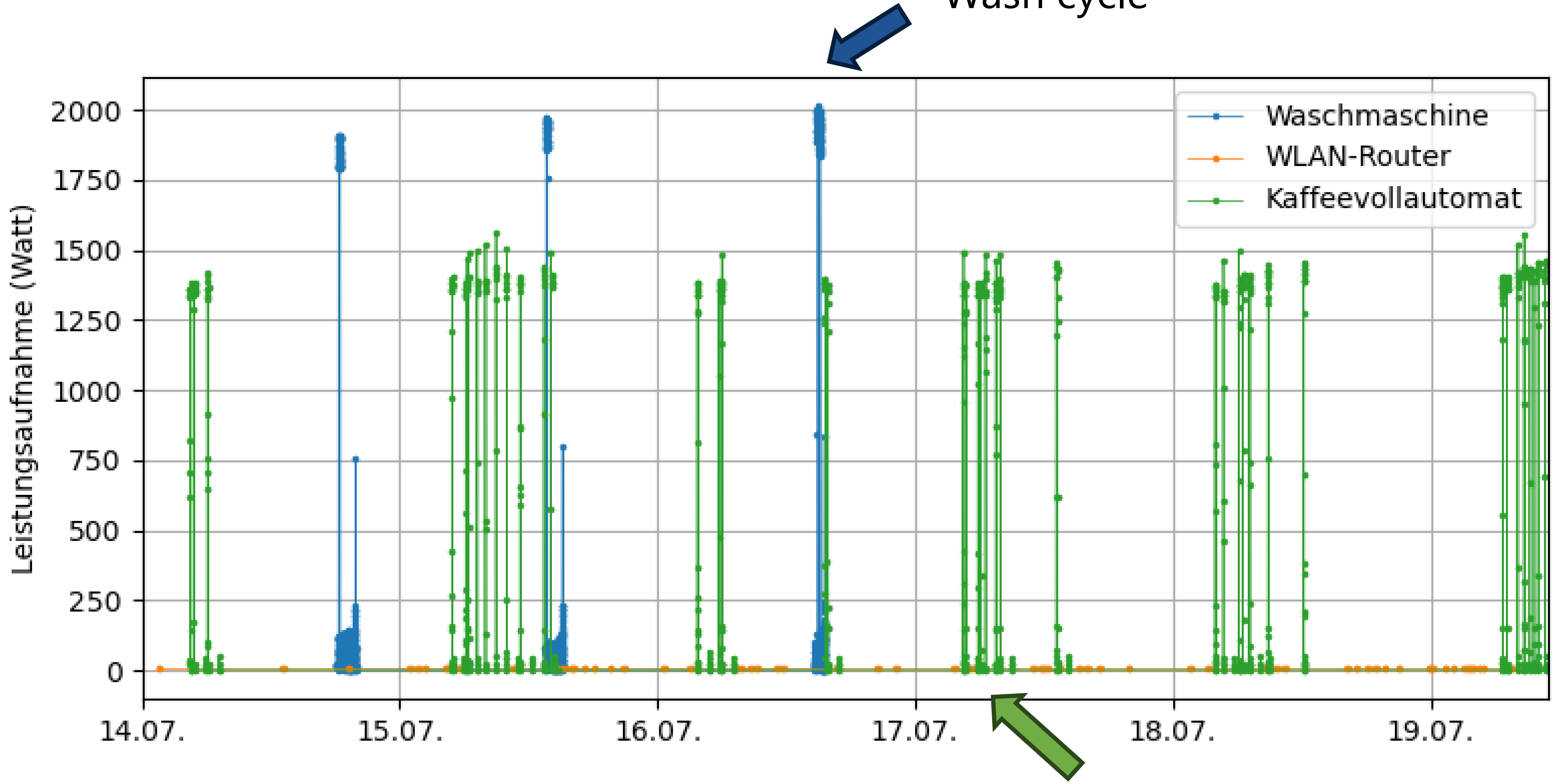




Conclusions and Discussions

Energy Consumption Observability & Metrics Needed

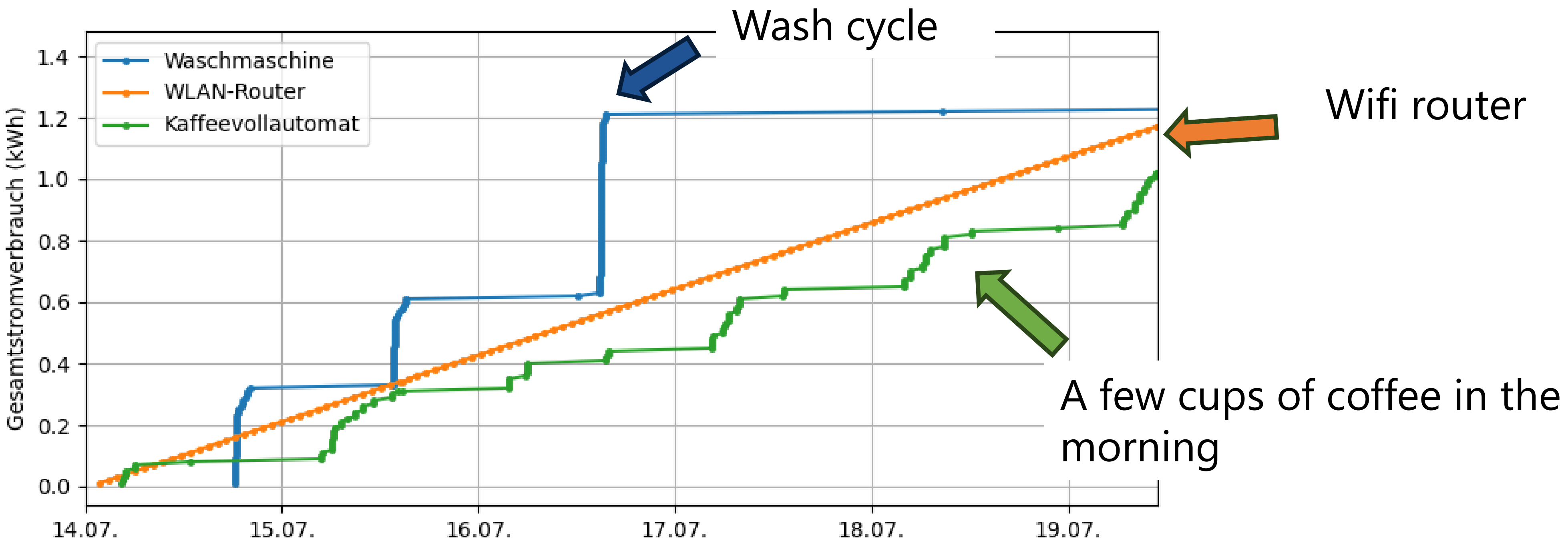
Wash cycle



A few cups of coffee in the morning

Energy Consumption Observability & Metrics Needed (f.)

- Total power consumption is the cumulative power drawn over time.



Energy Efficiency Metrics

- All measures have their meaning
- Single metric not sufficient to evaluate systems
- Which metric depends on use case, what is considered, etc.

Energy Efficiency

- **Absolute Measures**: Consumption Related to Output
- **Relative Measures (%):** Relation within System
- **Relative Measures (%):** (Ideal) Reference System

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Conclusions

- There is **no single KPI** that fully captures energy efficiency in mobile networks.
- Measuring only EI overlooks critical dimensions of network value and performance
- Metric needs to consider different use cases and scenarios: comparison, benchmarking, planning, operations, reporting

- The key question is not: "How many Joule per bit?"
- But: "How do we optimize cost, quality of service, and value delivered per unit of energy?"

- Energy efficiency is not just a technical metric: **system-level optimization** challenge balancing performance, QoE, cost, … (e.g. per user QoS/QoE metrics)

- Sustainability beyond energy efficiency: frequent switching on/off impacts **lifetime** of devices (embodied emissions, costs)

SUSTAINET advance

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Supporting Projects

- Green and Energy Monitoring for Future Network Infrastructures enabling also Large Deployments (GREENFIELD)
  - http://comnets.org/projects/greenfield/

- Sustainable Technologies for Advanced Resilient and Energy-Efficient Networks (SUSTAINET), CELTIC-NEXT Next project.
  - https://www.celticnext.eu/project-sustainet/

Thank you for your attention

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